

Stage 1, Stage 2 and Stage 3 Results

Dataset	Encoder	Stage	Method	T	K-shot	Runs	Test Accuracy	Delta	Baseline
dtd	dinov2_vits14	1	linear_probe	-	5	3	63.01% +/- 0.27%	-	-
dtd	dinov2_vits14	1	linear_probe	-	10	3	68.58% +/- 0.80%	-	-
dtd	dinov2_vits14	1	linear_probe	-	full	3	76.35% +/- 0.72%	-	-
dtd	dinov2_vits14	3	fm_cls_rolled	4	10	3	68.78% +/- 0.61%	+0.20% +/- 0.19%	linear_probe
dtd	dinov2_vits14	3	fm_cls_guided	4	10	3	69.63% +/- 0.37%	+1.05% +/- 0.87%	linear_probe
dtd	dinov2_vits14	1	prototype	-	5	3	64.33% +/- 0.30%	-	-
dtd	dinov2_vits14	1	prototype	-	10	3	68.46% +/- 1.01%	-	-
dtd	dinov2_vits14	1	prototype	-	full	1	72.29%	-	-
dtd	dinov2_vits14	2	fm_standard	4	5	3	61.31% +/- 0.75%	-3.01% +/- 0.46%	prototype
dtd	dinov2_vits14	2	fm_standard	12	5	3	61.70% +/- 0.83%	-2.62% +/- 0.55%	prototype
dtd	dinov2_vits14	2	fm_standard	4	10	3	65.60% +/- 0.59%	-2.85% +/- 1.45%	prototype
dtd	dinov2_vits14	2	fm_standard	12	10	3	65.92% +/- 0.57%	-2.54% +/- 1.48%	prototype
dtd	dinov2_vits14	2	fm_standard	4	full	1	74.63%	+2.34%	prototype
dtd	dinov2_vits14	2	fm_standard	12	full	1	74.79%	+2.50%	prototype
dtd	dinov2_vits14	2	fm_rolled	4	5	3	57.48% +/- 1.32%	-6.84% +/- 1.02%	prototype
dtd	dinov2_vits14	2	fm_rolled	12	5	3	57.50% +/- 1.04%	-6.83% +/- 0.73%	prototype
dtd	dinov2_vits14	2	fm_rolled	4	10	3	61.74% +/- 0.48%	-6.72% +/- 1.46%	prototype
dtd	dinov2_vits14	2	fm_rolled	12	10	3	61.74% +/- 0.27%	-6.72% +/- 1.28%	prototype
dtd	dinov2_vits14	2	fm_rolled	4	full	1	73.51%	+1.22%	prototype
dtd	dinov2_vits14	2	fm_rolled	12	full	1	72.93%	+0.64%	prototype
dtd	dinov2_vits14	3 ext	cls_finetune	4	10	3	68.90% +/- 0.80%	+0.32% +/- 0.00%	linear_probe
dtd	dinov2_vits14	3 ext	fm_cls_joint	4	10	3	69.41% +/- 0.35%	+0.83% +/- 0.46%	linear_probe
dtd	resnet18	1	linear_probe	-	5	3	45.80% +/- 0.75%	-	-
dtd	resnet18	1	linear_probe	-	10	3	52.39% +/- 0.74%	-	-
dtd	resnet18	1	linear_probe	-	full	3	62.55% +/- 0.51%	-	-
dtd	resnet18	1	prototype	-	5	3	46.93% +/- 0.32%	-	-

Dataset	Encoder	Stage	Method	T	K-shot	Runs	Test Accuracy	Delta	Baseline
dtd	resnet18	1	prototype	-	10	3	51.97% +/- 0.54%	-	-
dtd	resnet18	1	prototype	-	full	1	58.78%	-	-
dtd	resnet18	2	fm_standard	4	5	3	44.82% +/- 1.45%	-2.11% +/- 1.19%	prototype
dtd	resnet18	2	fm_standard	12	5	3	45.14% +/- 1.36%	-1.79% +/- 1.07%	prototype
dtd	resnet18	2	fm_standard	4	10	3	48.65% +/- 1.02%	-3.32% +/- 0.52%	prototype
dtd	resnet18	2	fm_standard	12	10	3	49.13% +/- 1.17%	-2.84% +/- 0.65%	prototype
dtd	resnet18	2	fm_standard	4	full	1	59.52%	+0.74%	prototype
dtd	resnet18	2	fm_standard	12	full	1	59.57%	+0.80%	prototype
dtd	resnet18	2	fm_rolled	4	5	3	40.51% +/- 0.75%	-6.42% +/- 0.45%	prototype
dtd	resnet18	2	fm_rolled	12	5	3	40.60% +/- 0.85%	-6.33% +/- 0.53%	prototype
dtd	resnet18	2	fm_rolled	4	10	3	40.62% +/- 1.00%	-11.35% +/- 1.20%	prototype
dtd	resnet18	2	fm_rolled	12	10	3	41.65% +/- 0.90%	-10.32% +/- 0.88%	prototype
dtd	resnet18	2	fm_rolled	4	full	1	56.06%	-2.71%	prototype
dtd	resnet18	2	fm_rolled	12	full	1	56.01%	-2.77%	prototype
flowers102	resnet18	1	linear_probe	-	5	3	75.13% +/- 0.22%	-	-
flowers102	resnet18	1	linear_probe	-	10	3	83.22% +/- 0.14%	-	-
flowers102	resnet18	1	linear_probe	-	full	3	83.22% +/- 0.14%	-	-
flowers102	resnet18	3	fm_cls_rolled	4	10	3	84.15% +/- 0.06%	+0.93% +/- 0.08%	linear_probe
flowers102	resnet18	3	fm_cls_guided	4	10	3	84.24% +/- 0.36%	+1.02% +/- 0.34%	linear_probe
flowers102	resnet18	1	prototype	-	5	3	69.62% +/- 0.32%	-	-
flowers102	resnet18	1	prototype	-	10	3	75.22% +/- 0.00%	-	-
flowers102	resnet18	1	prototype	-	full	1	75.22%	-	-
flowers102	resnet18	2	fm_standard	4	5	3	69.19% +/- 1.02%	-0.43% +/- 0.85%	prototype
flowers102	resnet18	2	fm_standard	12	5	3	69.59% +/- 1.01%	-0.02% +/- 0.82%	prototype
flowers102	resnet18	2	fm_standard	4	10	3	76.83% +/- 0.11%	+1.62% +/- 0.11%	prototype
flowers102	resnet18	2	fm_standard	12	10	3	77.17% +/- 0.21%	+1.95% +/- 0.21%	prototype
flowers102	resnet18	2	fm_standard	4	full	1	76.79%	+1.58%	prototype
flowers102	resnet18	2	fm_standard	12	full	1	76.94%	+1.72%	prototype

Dataset	Encoder	Stage	Method	T	K-shot	Runs	Test Accuracy	Delta	Baseline
flowers102	resnet18	2	fm_rolled	4	5	3	62.54% +/- 1.32%	-7.07% +/- 1.12%	prototype
flowers102	resnet18	2	fm_rolled	12	5	3	63.07% +/- 0.68%	-6.54% +/- 0.40%	prototype
flowers102	resnet18	2	fm_rolled	4	10	3	72.46% +/- 0.38%	-2.76% +/- 0.38%	prototype
flowers102	resnet18	2	fm_rolled	12	10	3	72.65% +/- 0.37%	-2.57% +/- 0.37%	prototype
flowers102	resnet18	2	fm_rolled	4	full	1	72.89%	-2.33%	prototype
flowers102	resnet18	2	fm_rolled	12	full	1	73.07%	-2.15%	prototype
flowers102	resnet18	3 ext	cls_finetune	4	10	3	83.43% +/- 0.07%	+0.22% +/- 0.17%	linear_probe
flowers102	resnet18	3 ext	fm_cls_joint	4	10	3	84.22% +/- 0.09%	+1.00% +/- 0.14%	linear_probe

Stage 3: Stage 1 linear probe vs. both Stage 3 strategies

Dataset	Encoder	Method	Test accuracy	Change vs. Stage 1 linear probe
dtd	dinov2_vits14	Stage 1 linear probe (baseline)	68.58% +/- 0.80%	-
dtd	dinov2_vits14	Strategy 1: end-to-end rolled-out (fm_cls_rolled)	68.78% +/- 0.61%	+0.20% +/- 0.19%
dtd	dinov2_vits14	Strategy 2: classifier-guided FM (fm_cls_guided)	69.63% +/- 0.37%	+1.05% +/- 0.87%
flowers102	resnet18	Stage 1 linear probe (baseline)	83.22% +/- 0.14%	-
flowers102	resnet18	Strategy 1: end-to-end rolled-out (fm_cls_rolled)	84.15% +/- 0.06%	+0.93% +/- 0.08%
flowers102	resnet18	Strategy 2: classifier-guided FM (fm_cls_guided)	84.24% +/- 0.36%	+1.02% +/- 0.34%

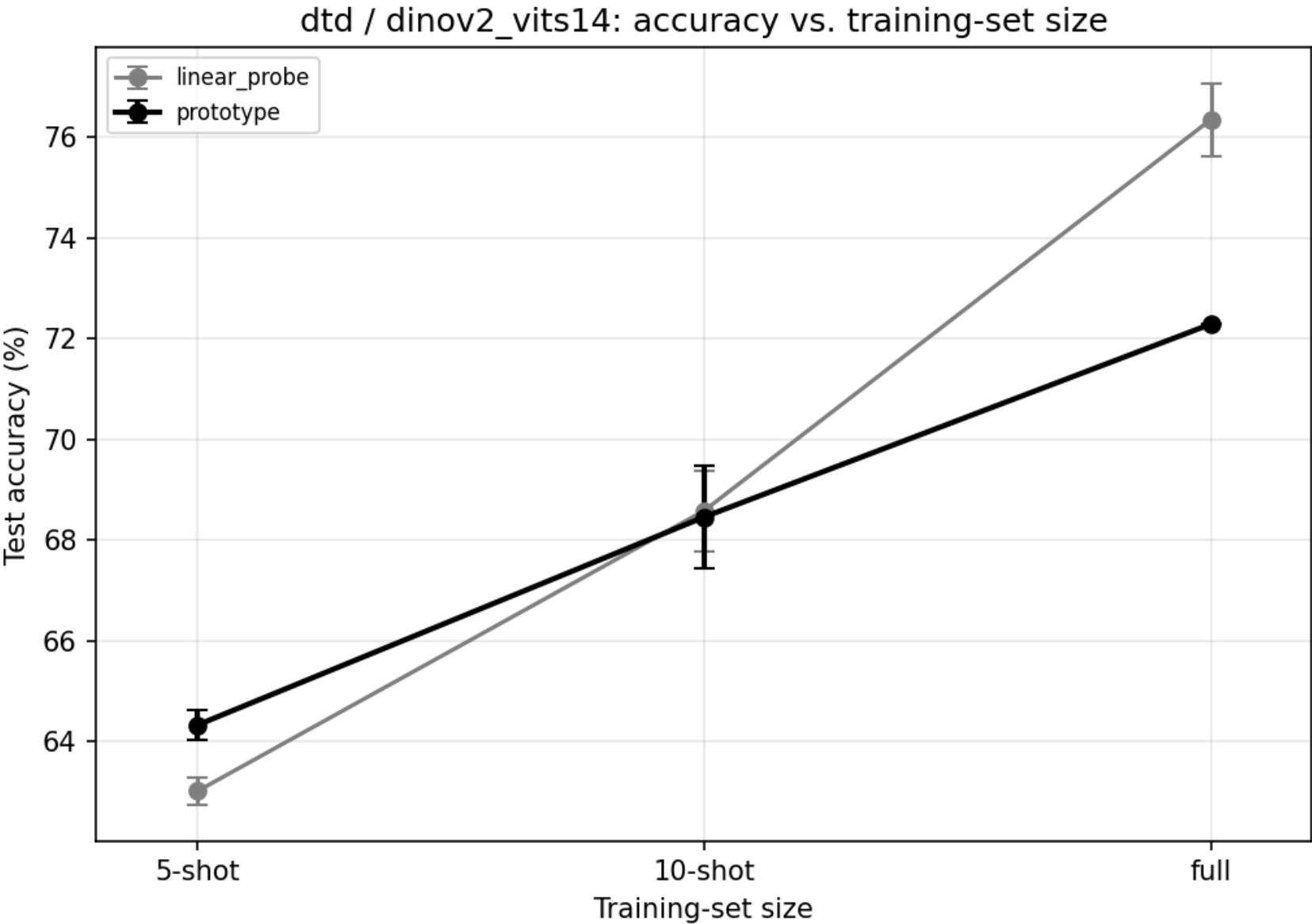
Stage 3 optional extension: jointly fine-tuning the classifier

Dataset	Encoder	Method	Test accuracy	Change vs. Stage 1 linear probe
dtd	dinov2_vits14	Stage 1 linear probe (baseline)	68.58% +/- 0.80%	-
dtd	dinov2_vits14	Control: classifier fine-tuned alone (cls_finetune)	68.90% +/- 0.80%	+0.32% +/- 0.00%
dtd	dinov2_vits14	Strategy 1: end-to-end rolled-out (fm_cls_rolled)	68.78% +/- 0.61%	+0.20% +/- 0.19%
dtd	dinov2_vits14	Extension: joint FM + classifier (fm_cls_joint)	69.41% +/- 0.35%	+0.83% +/- 0.46%
flowers102	resnet18	Stage 1 linear probe (baseline)	83.22% +/- 0.14%	-
flowers102	resnet18	Control: classifier fine-tuned alone (cls_finetune)	83.43% +/- 0.07%	+0.22% +/- 0.17%

Dataset	Encoder	Method	Test accuracy	Change vs. Stage 1 linear probe
flowers102	resnet18	Strategy 1: end-to-end rolled-out (fm_cls_rolled)	84.15% +/- 0.06%	+0.93% +/- 0.08%
flowers102	resnet18	Extension: joint FM + classifier (fm_cls_joint)	84.22% +/- 0.09%	+1.00% +/- 0.14%

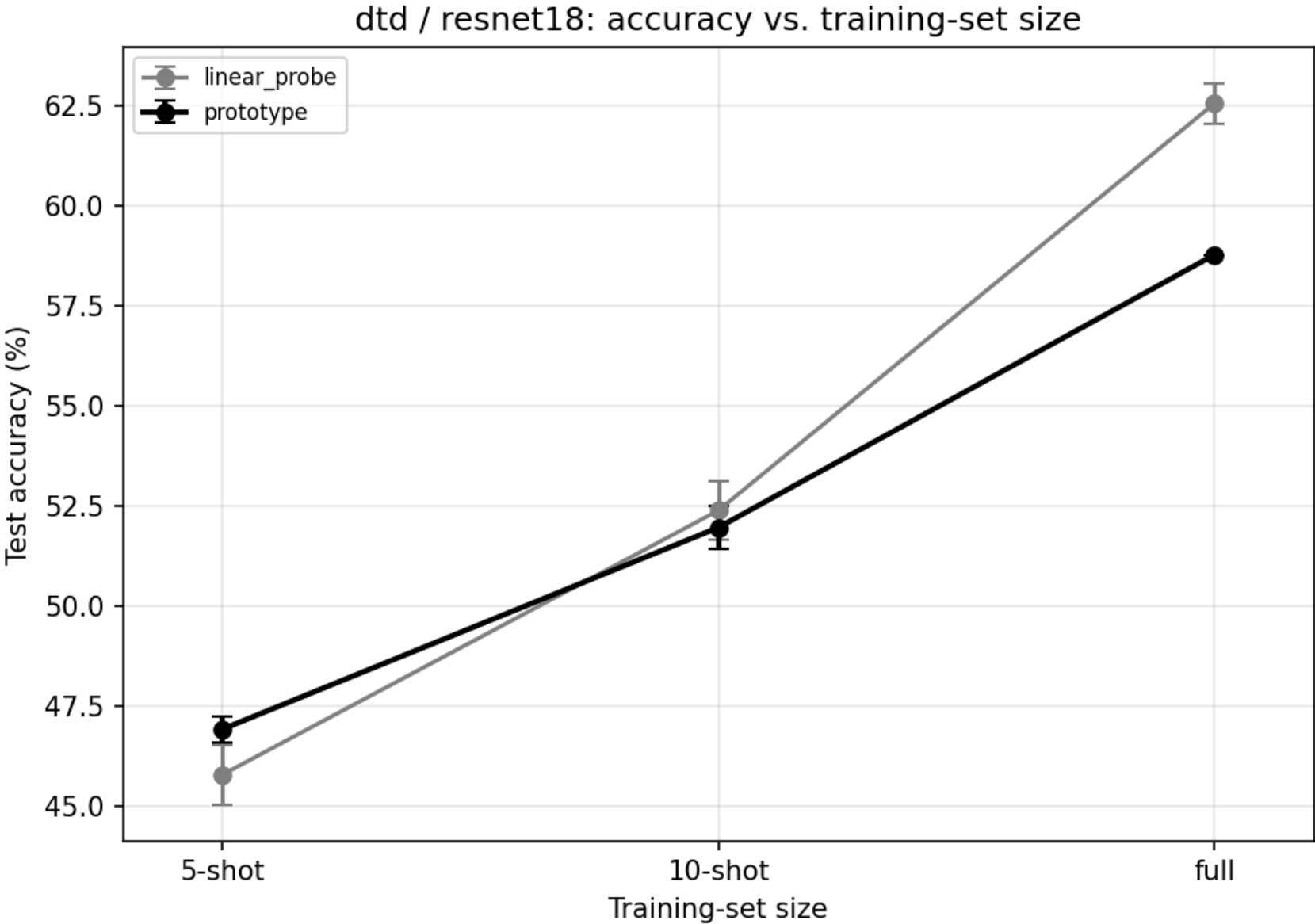
Accuracy vs. training-set size

dtd / dinov2_vits14



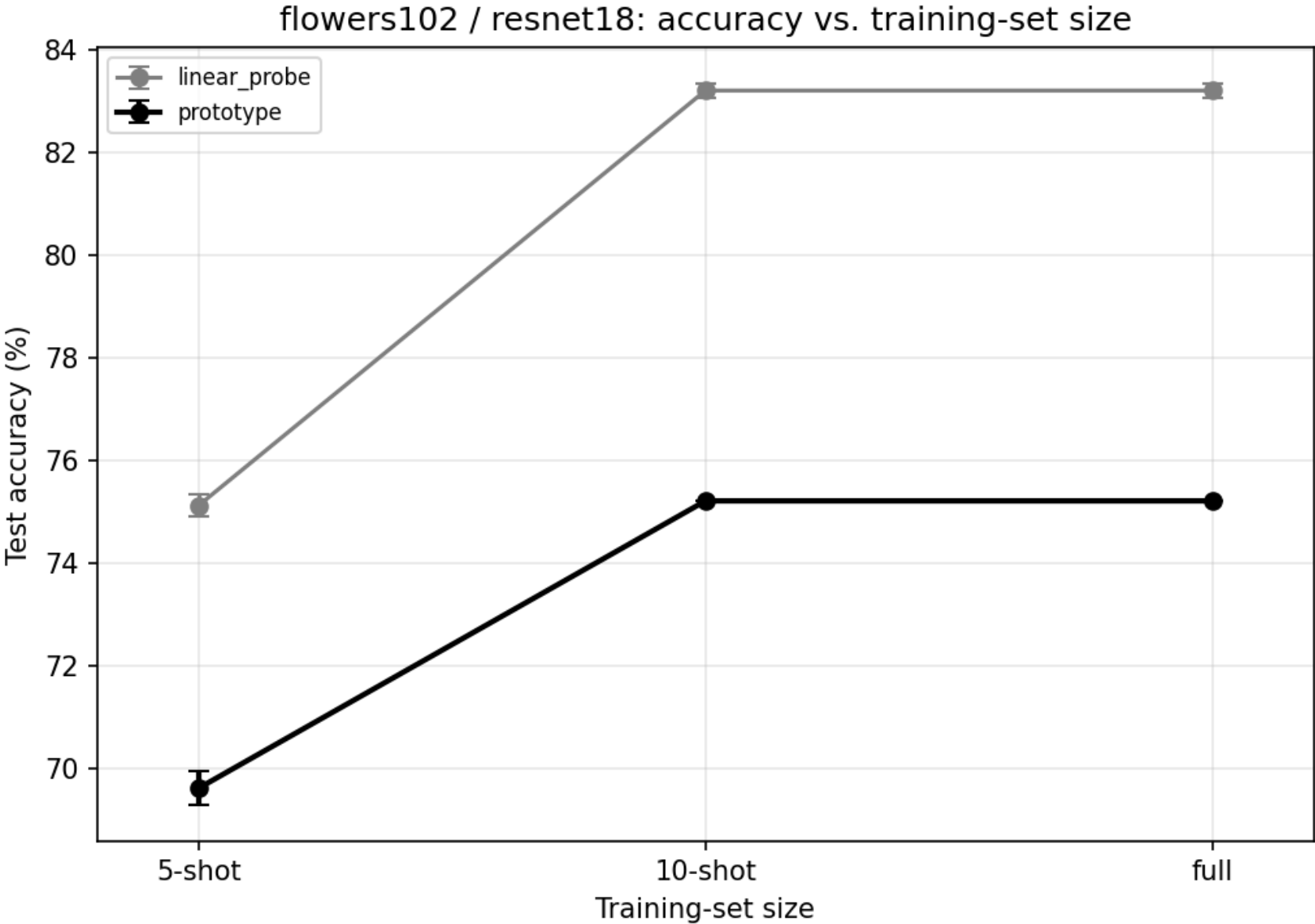
Accuracy vs. training-set size

dtd / resnet18



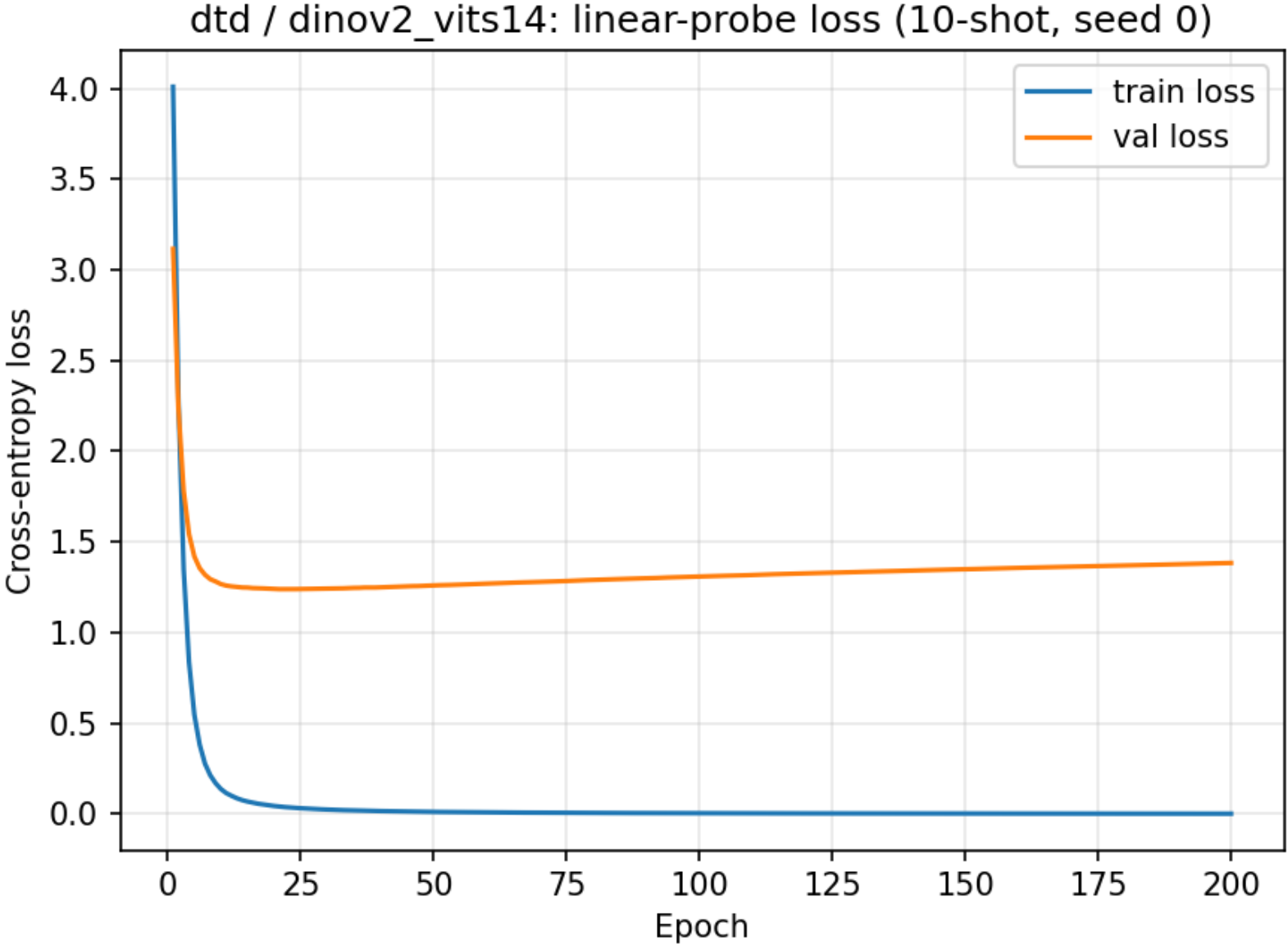
Accuracy vs. training-set size

flowers102 / resnet18



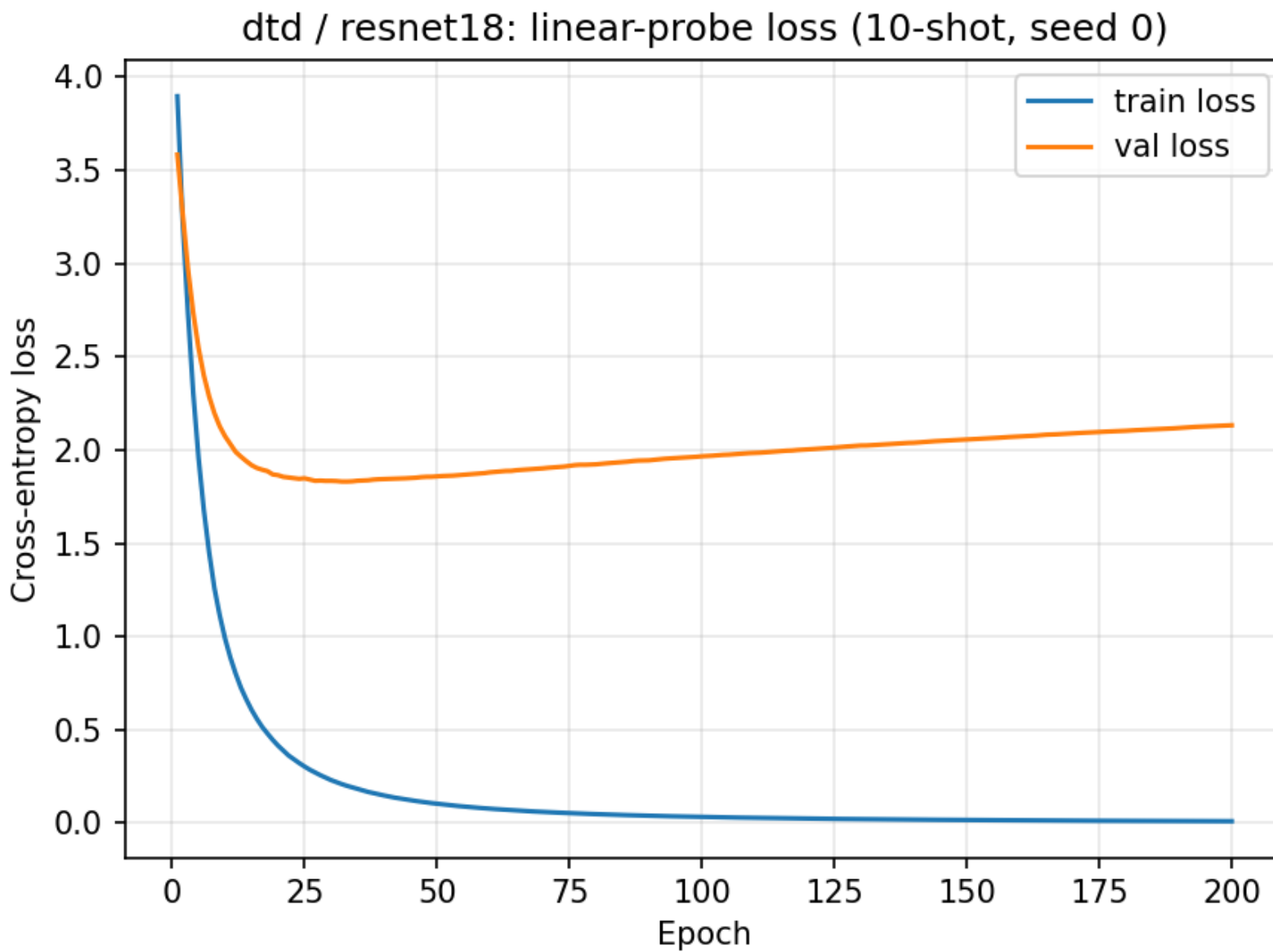
Training / validation loss (10-shot, seed 0)

dtd / dinov2_vits14



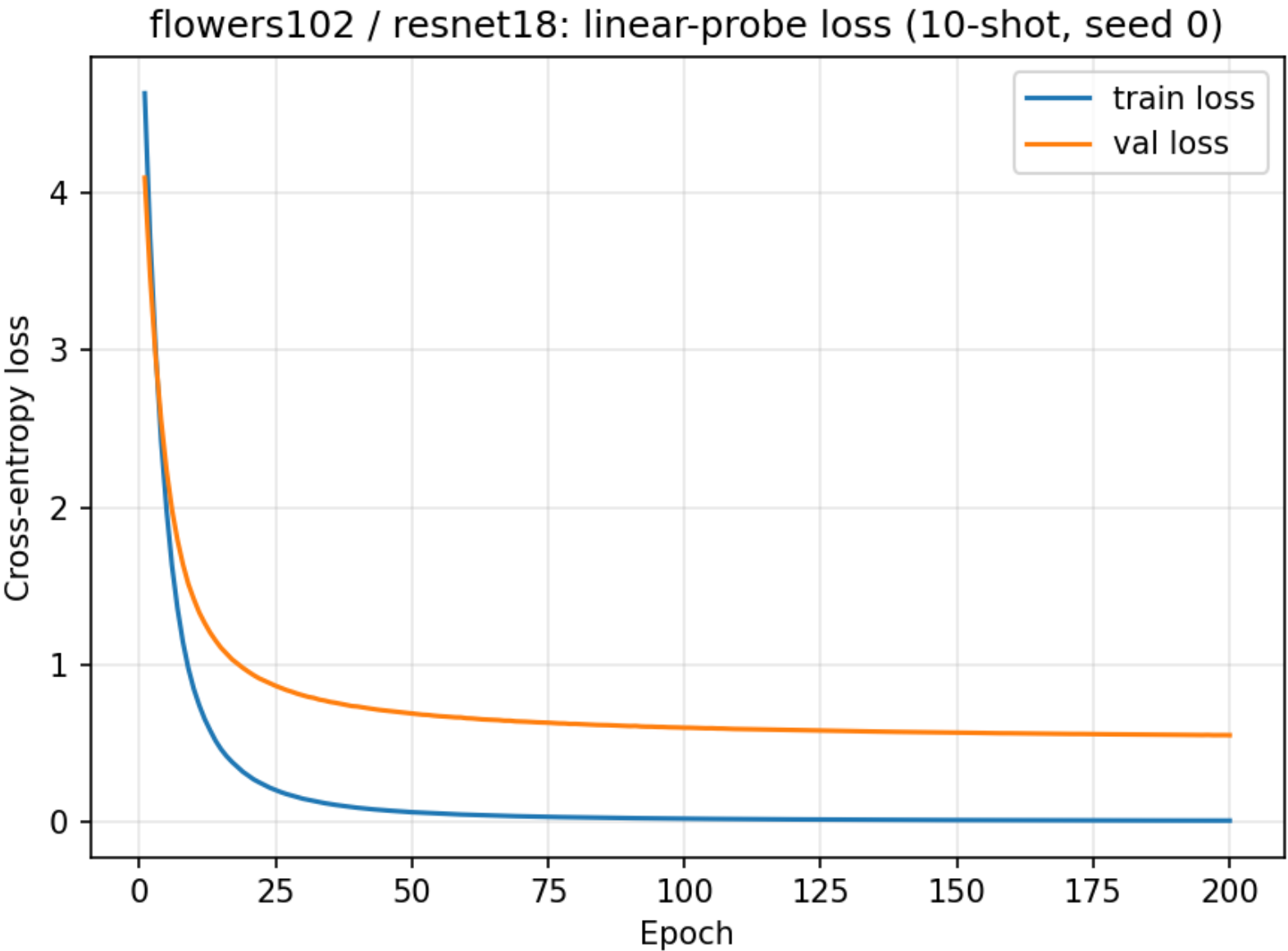
Training / validation loss (10-shot, seed 0)

dtd / resnet18



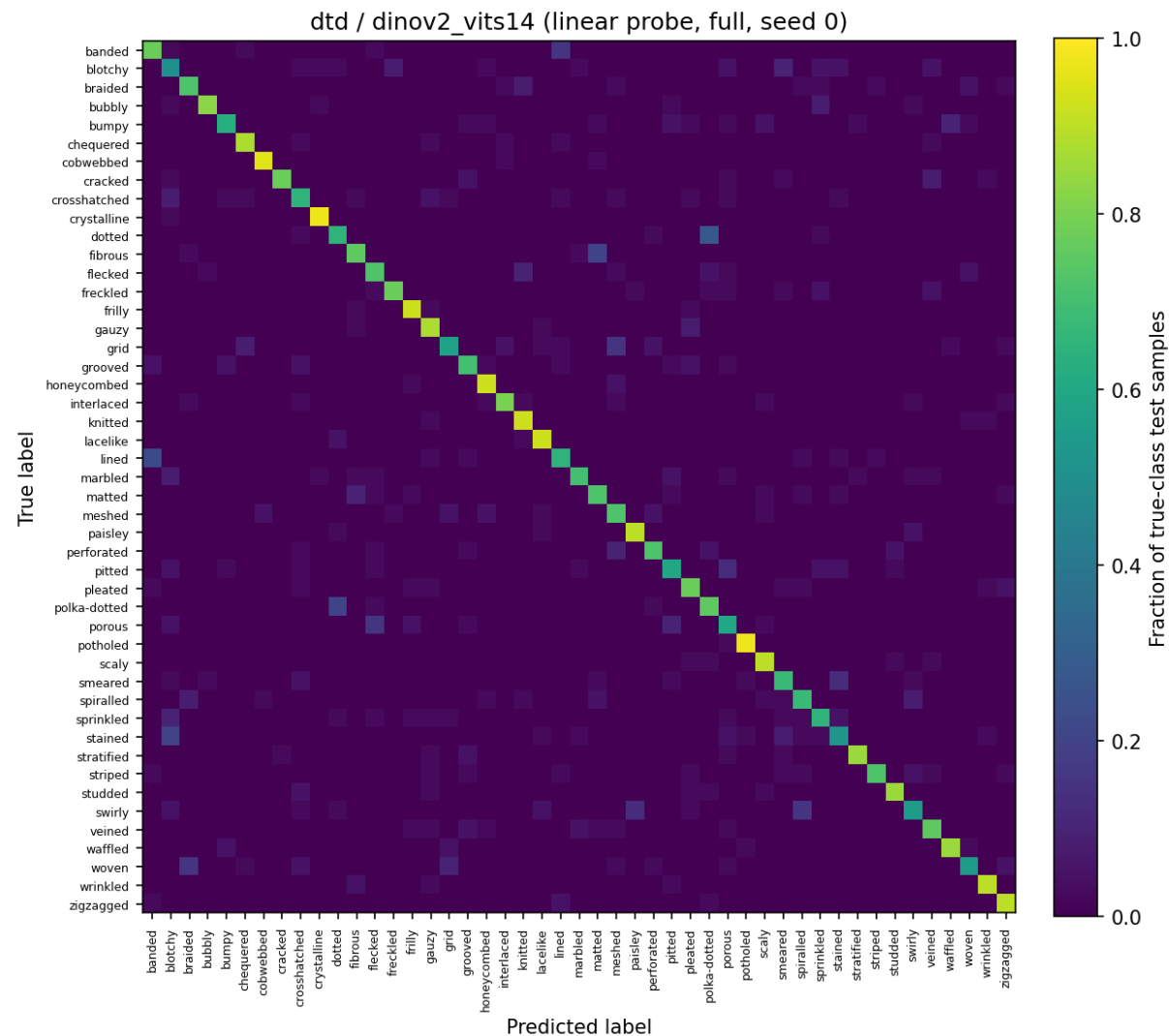
Training / validation loss (10-shot, seed 0)

flowers102 / resnet18



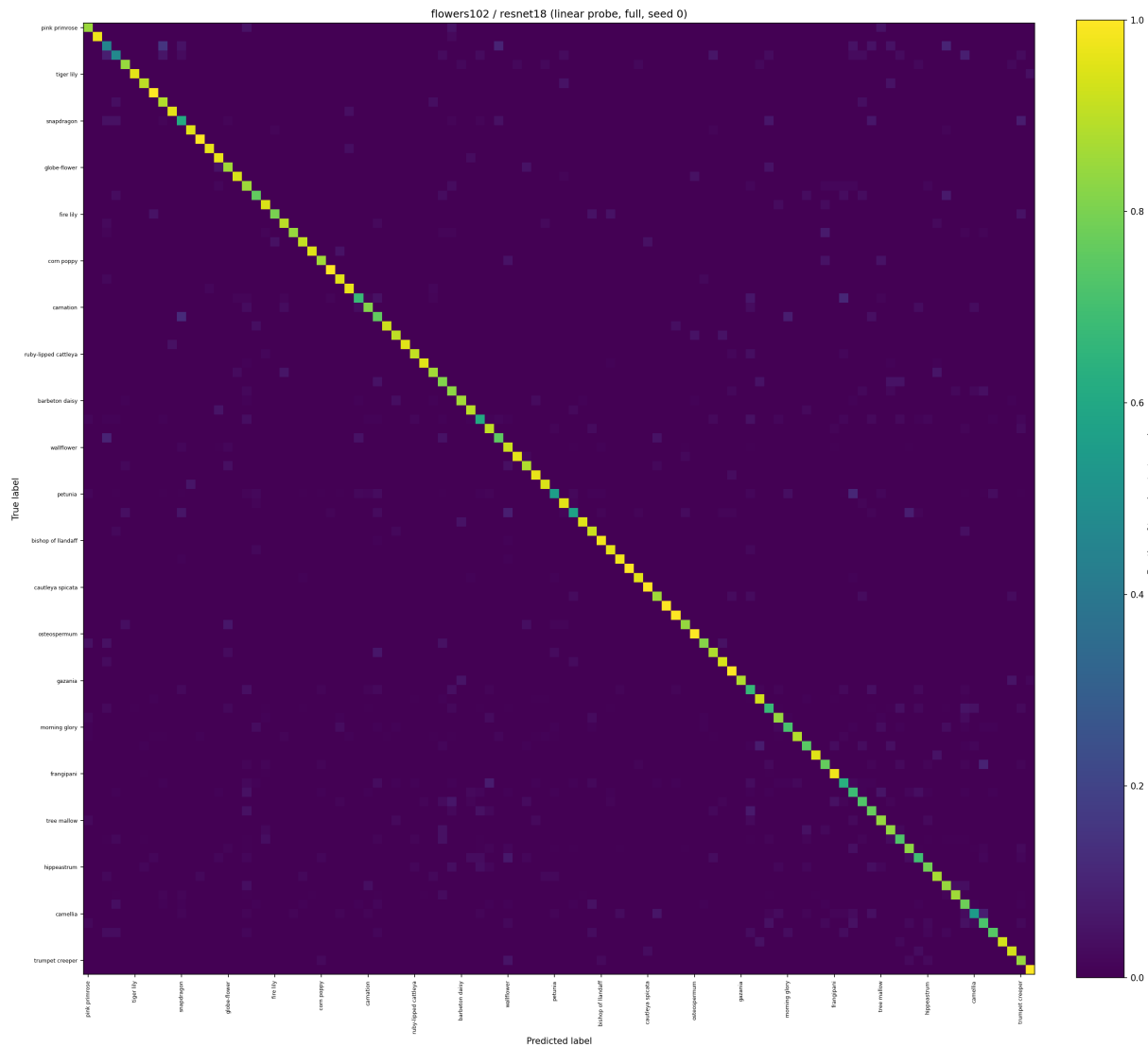
Confusion matrices (row-normalized)

dtd / dinov2_vits14



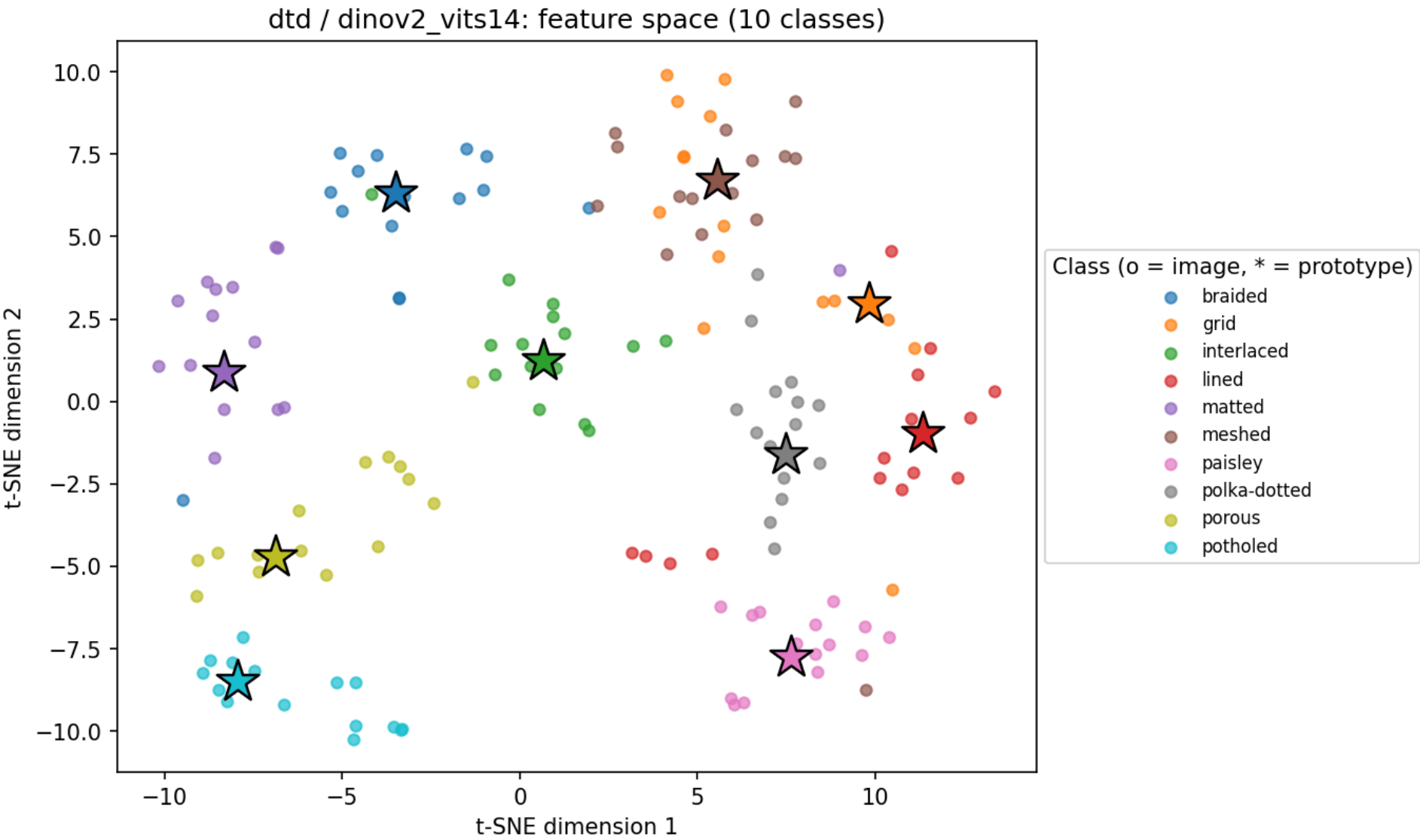
Confusion matrices (row-normalized)

flowers102 / resnet18



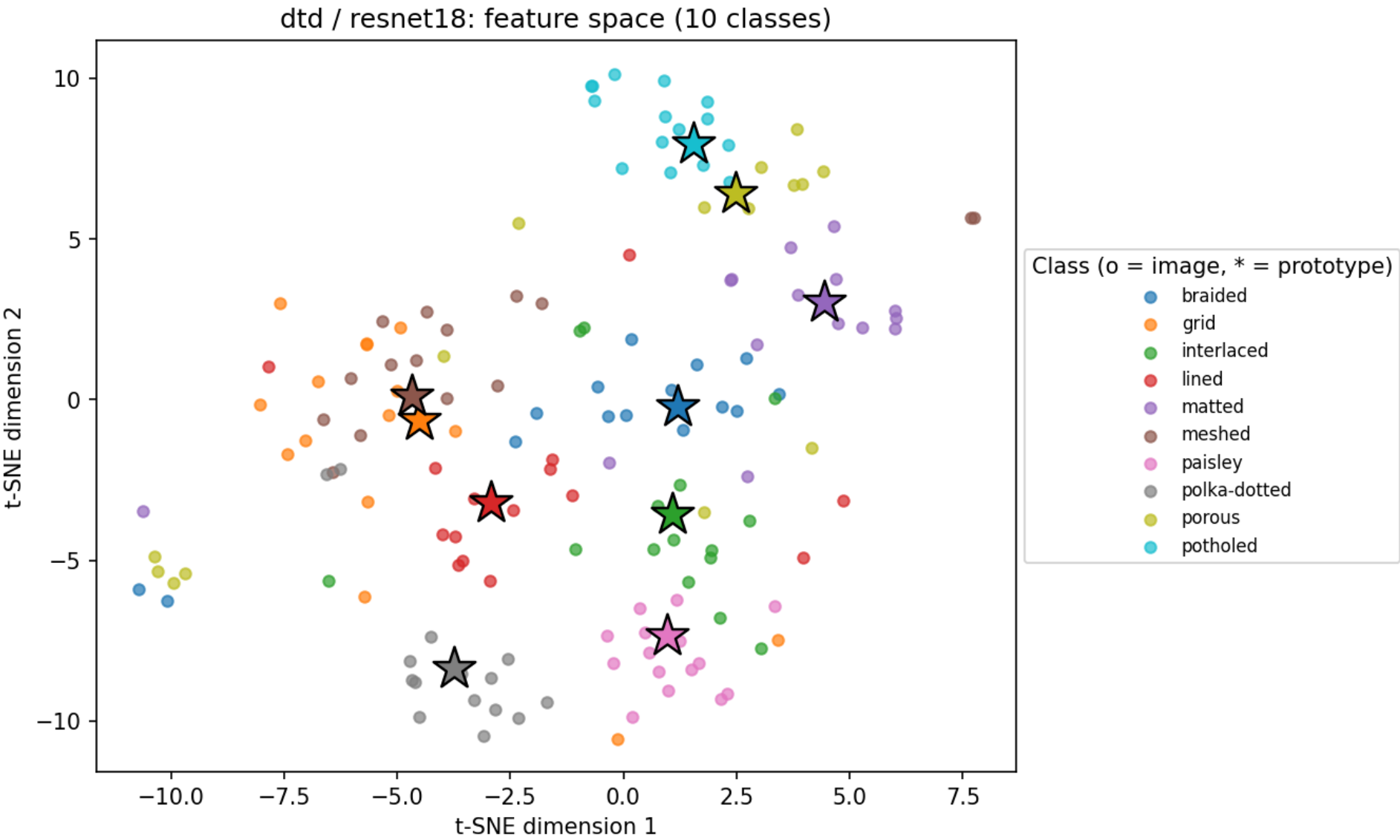
Feature-space visualizations (t-SNE)

dtd / dinov2_vits14



Feature-space visualizations (t-SNE)

dtd / resnet18



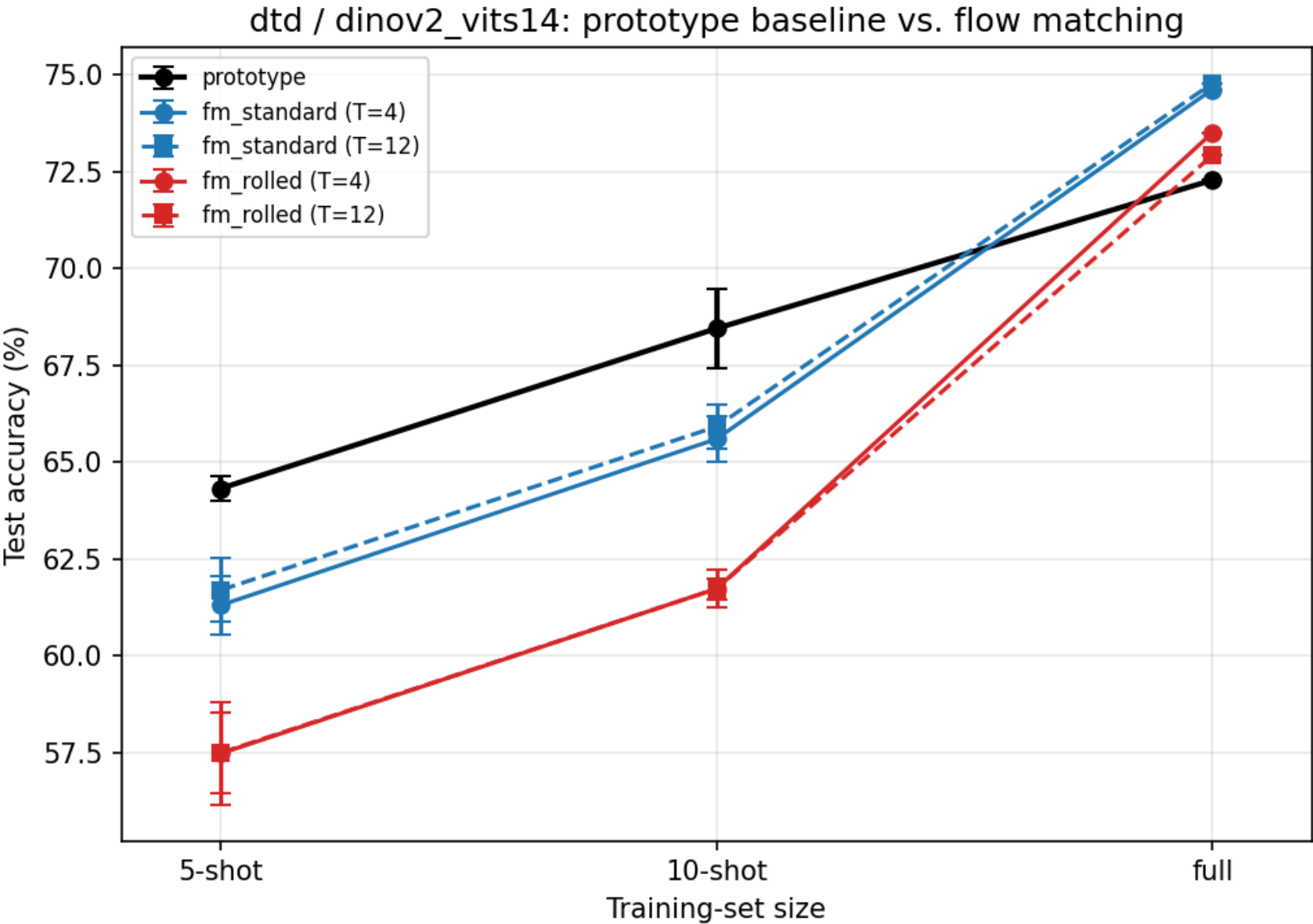
Feature-space visualizations (t-SNE)

flowers102 / resnet18



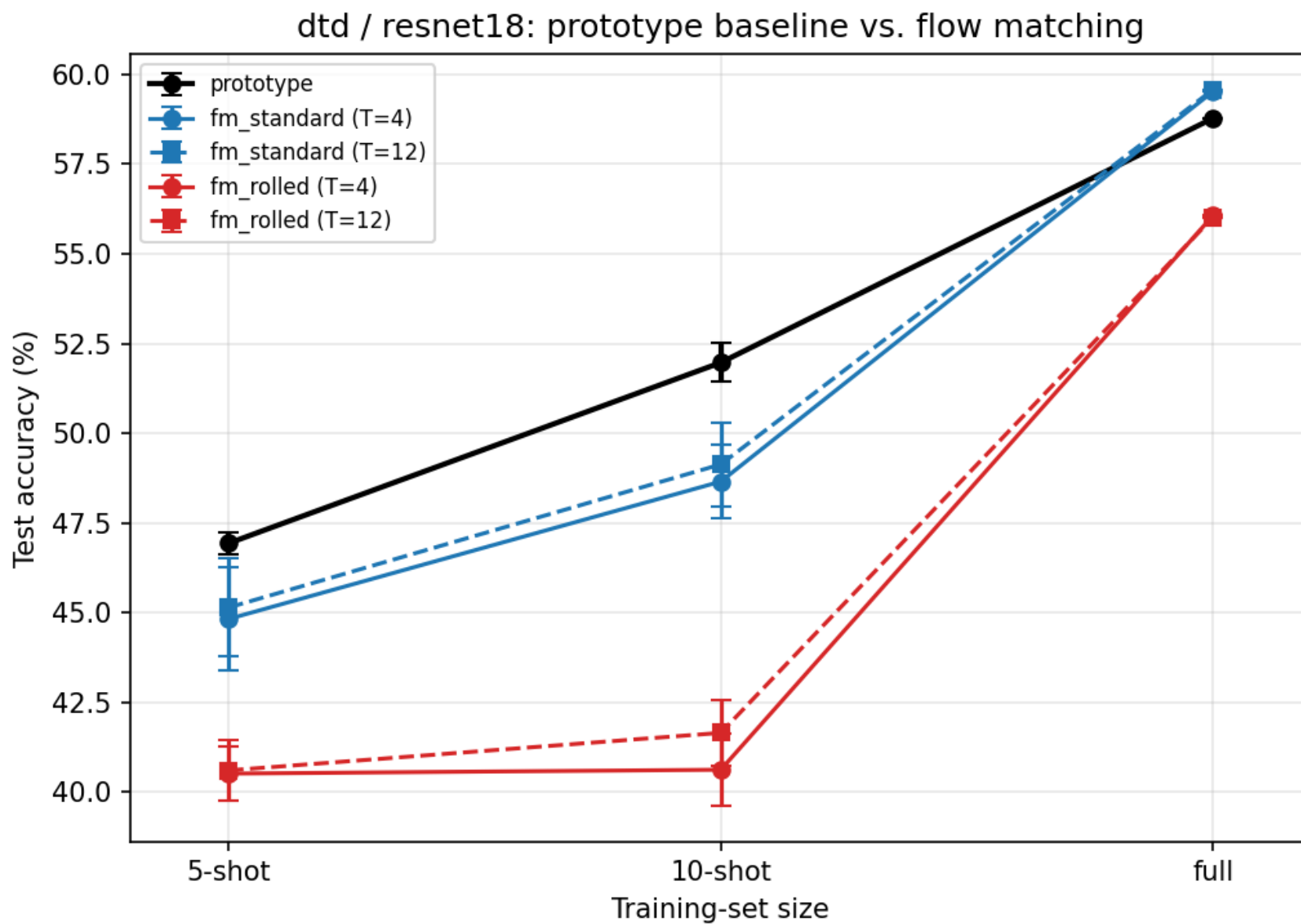
Stage 2: accuracy vs. training-set size

dtd / dinov2_vits14



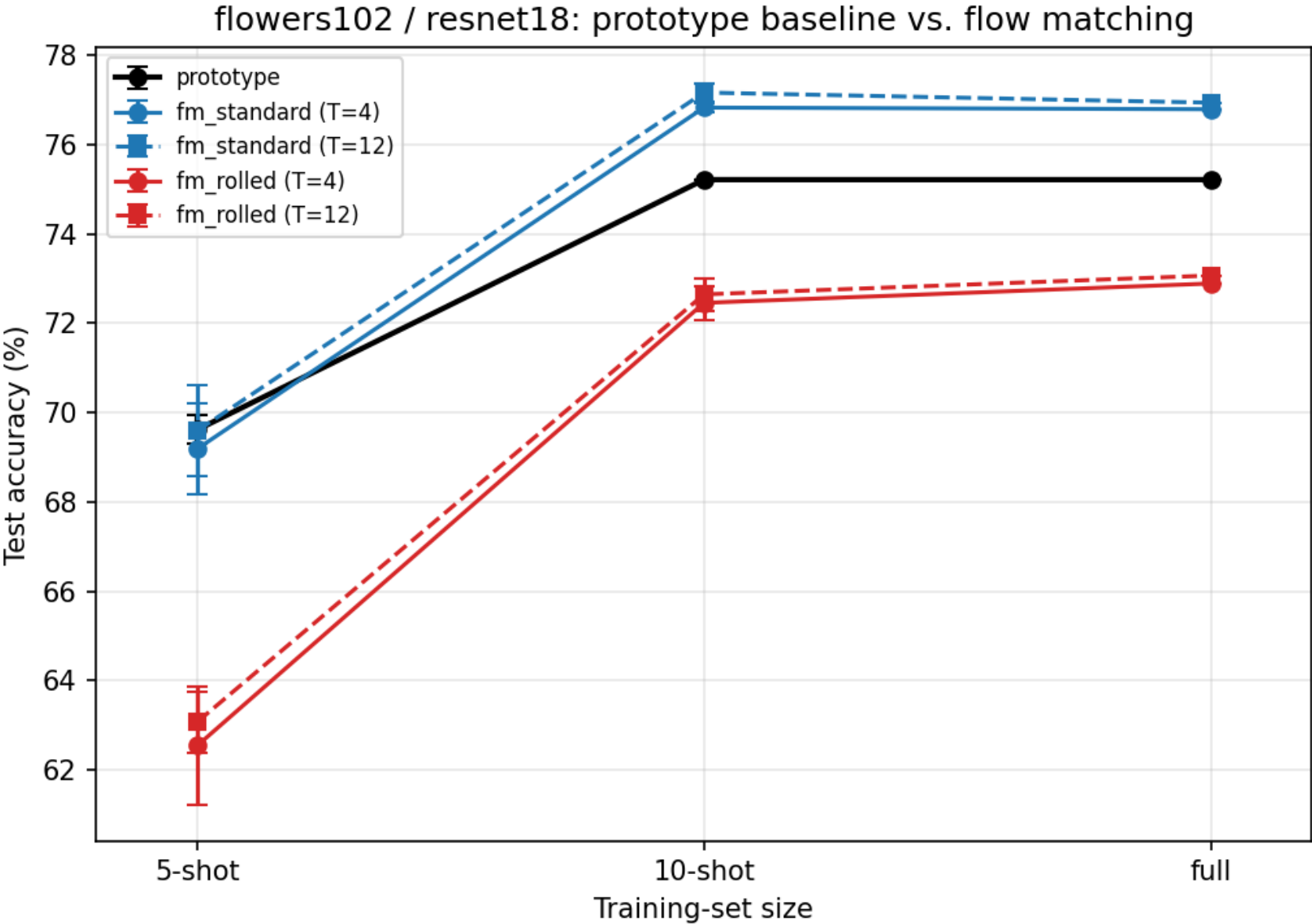
Stage 2: accuracy vs. training-set size

dtd / resnet18



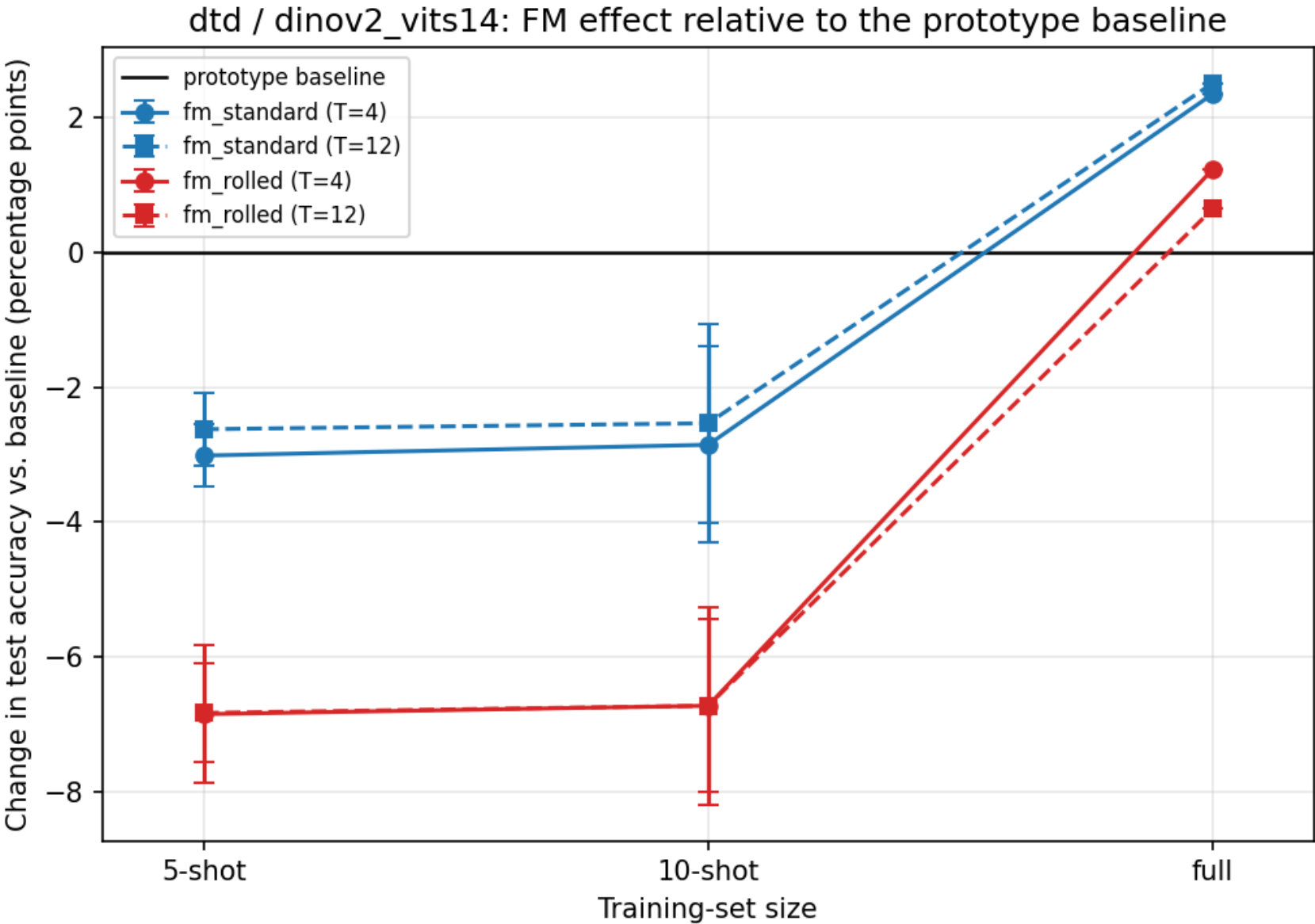
Stage 2: accuracy vs. training-set size

flowers102 / resnet18



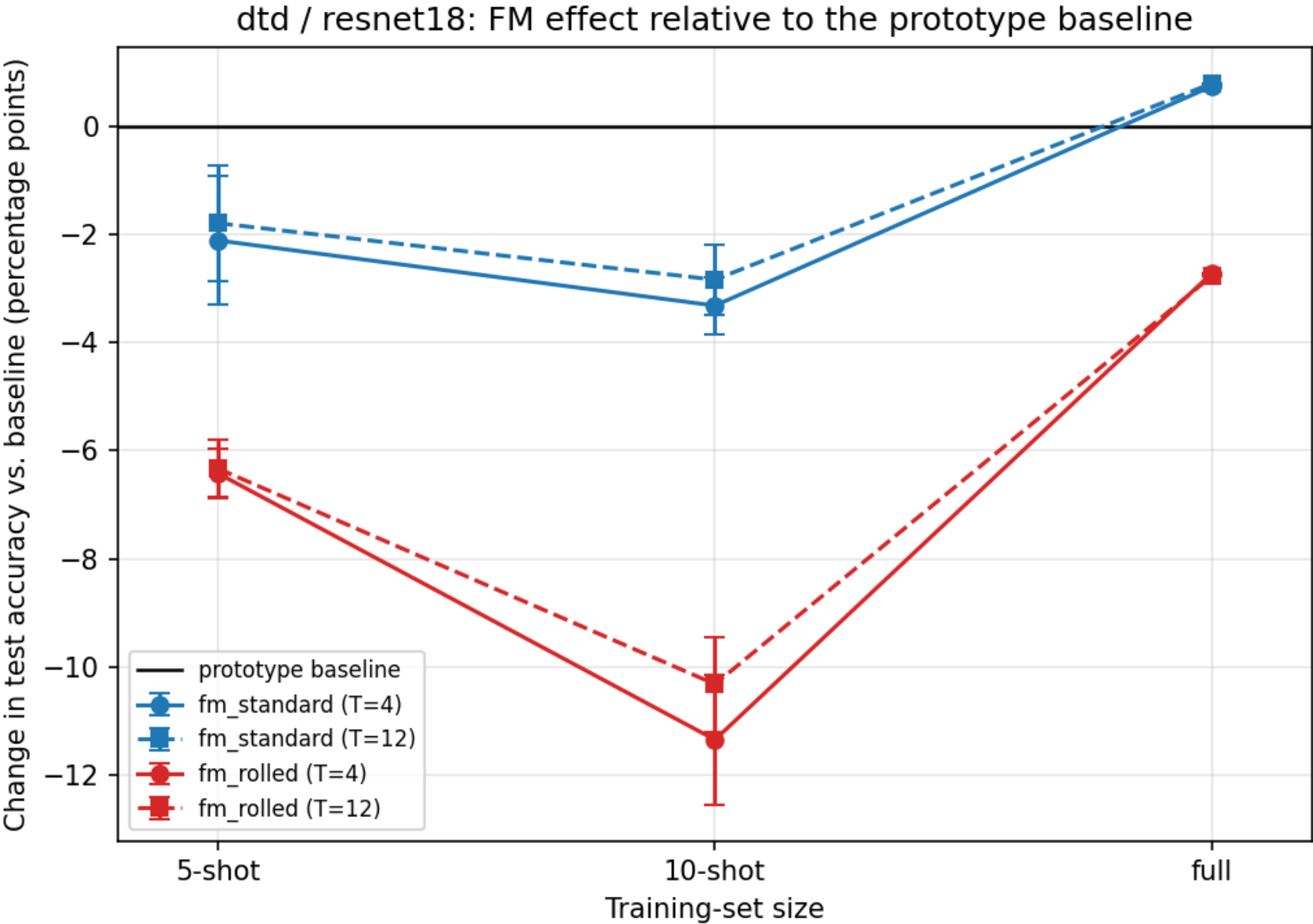
Stage 2: change in accuracy relative to the prototype baseline

dtd / dinov2_vits14



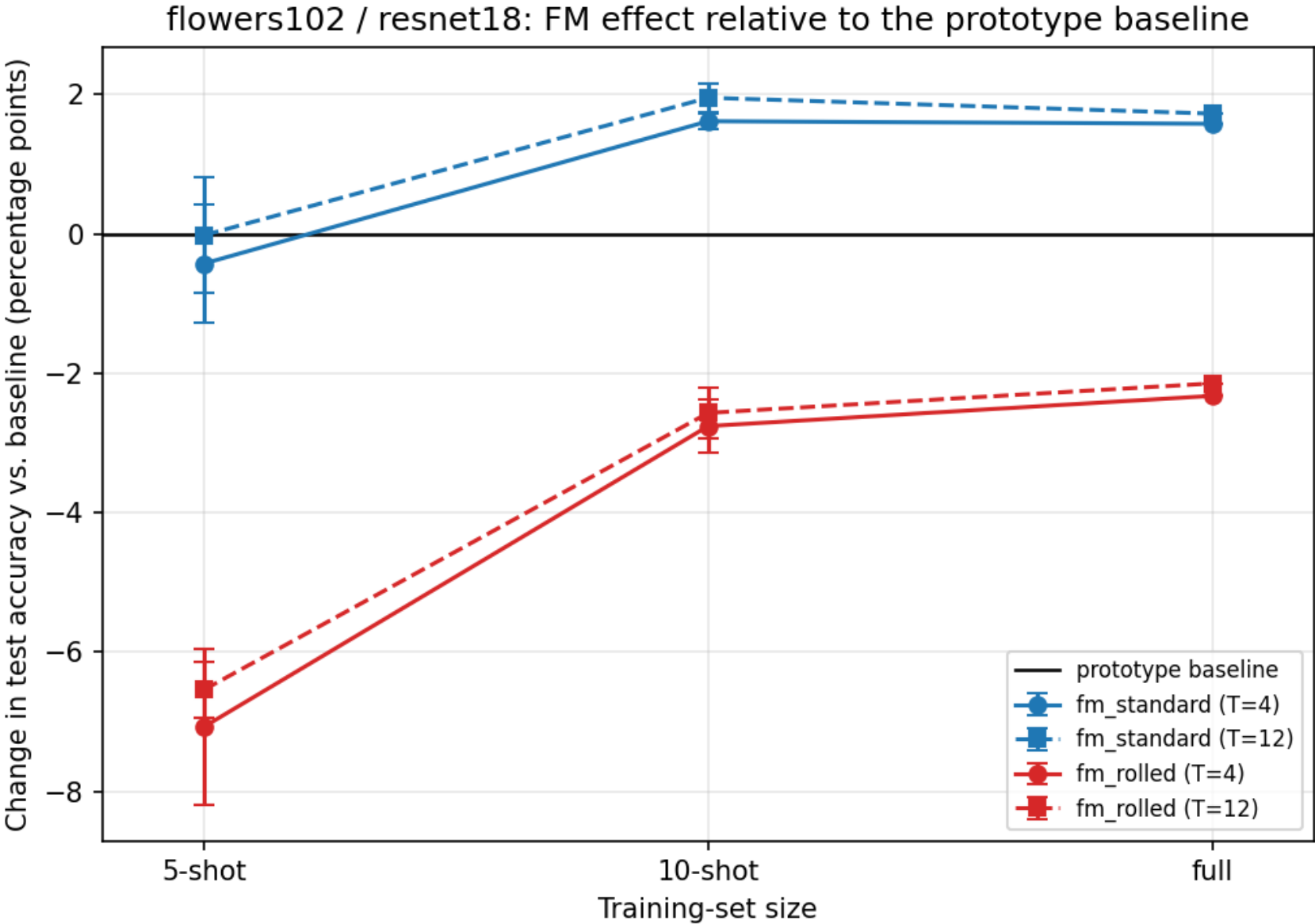
Stage 2: change in accuracy relative to the prototype baseline

dtd / resnet18



Stage 2: change in accuracy relative to the prototype baseline

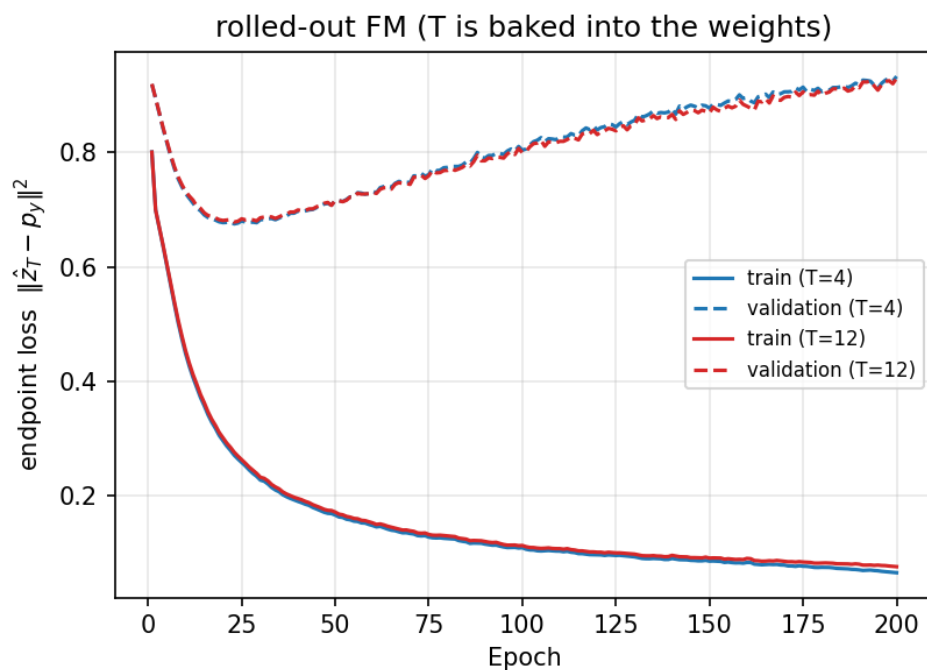
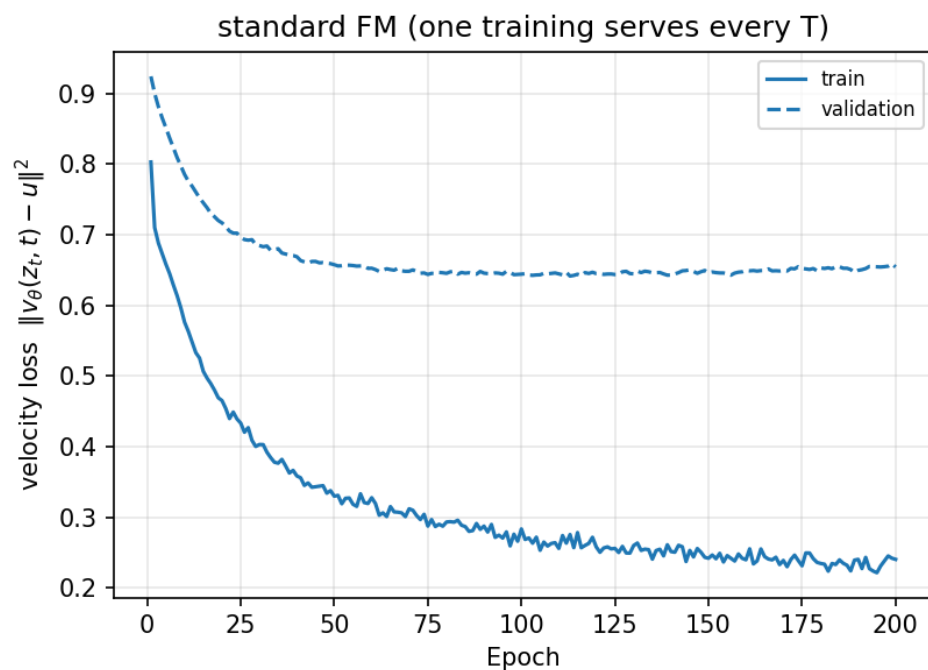
flowers102 / resnet18



Stage 2: flow-matching training curves (10-shot, seed 0)

dtd / dinov2_vits14

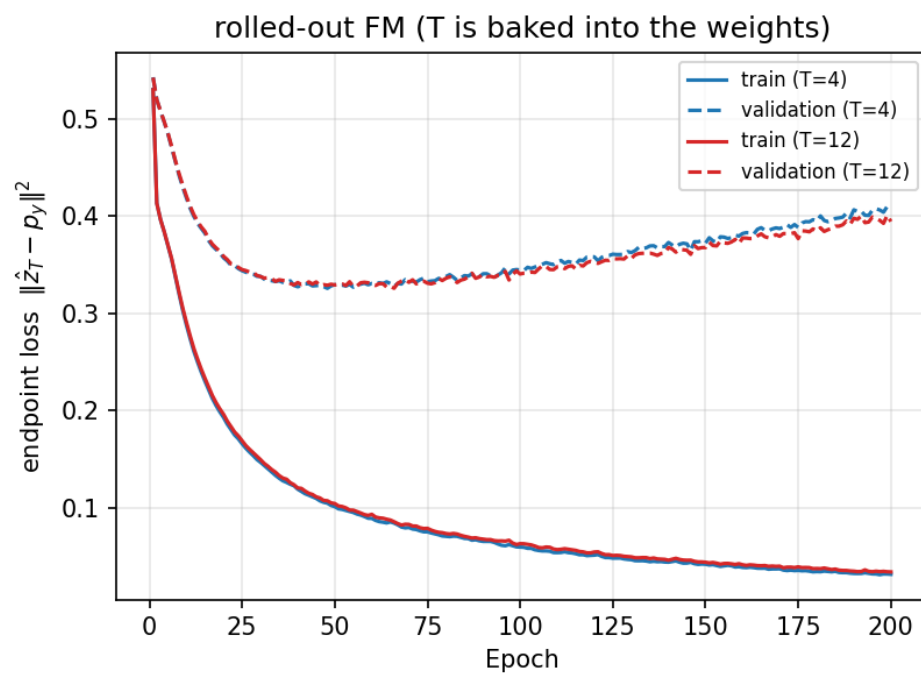
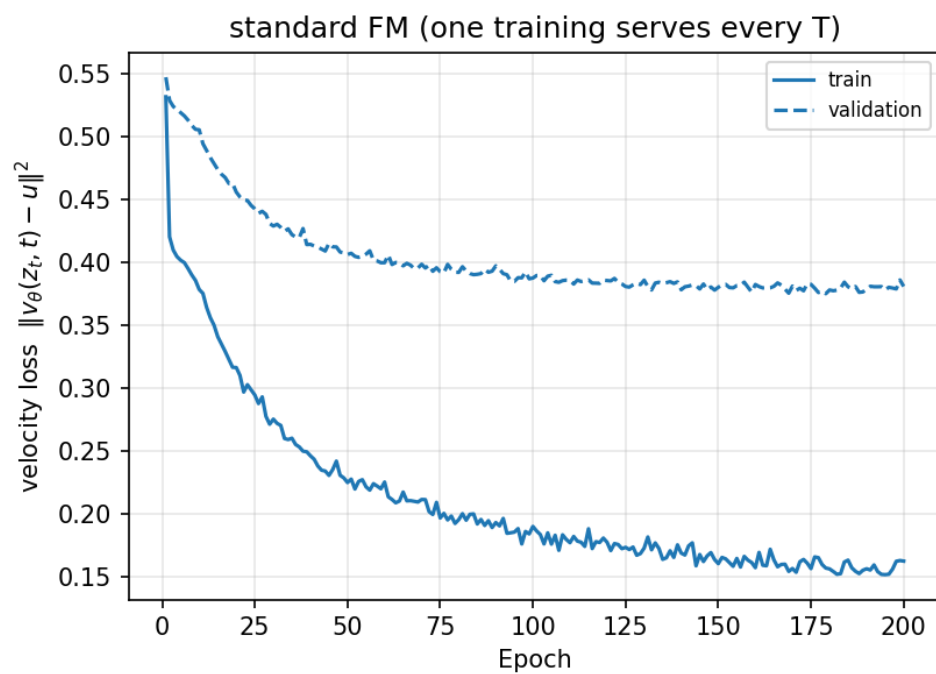
dtd / dinov2_vits14: flow-matching training (10-shot, seed 0)
note: the two panels measure different quantities and are not comparable to each other



Stage 2: flow-matching training curves (10-shot, seed 0)

dtd / resnet18

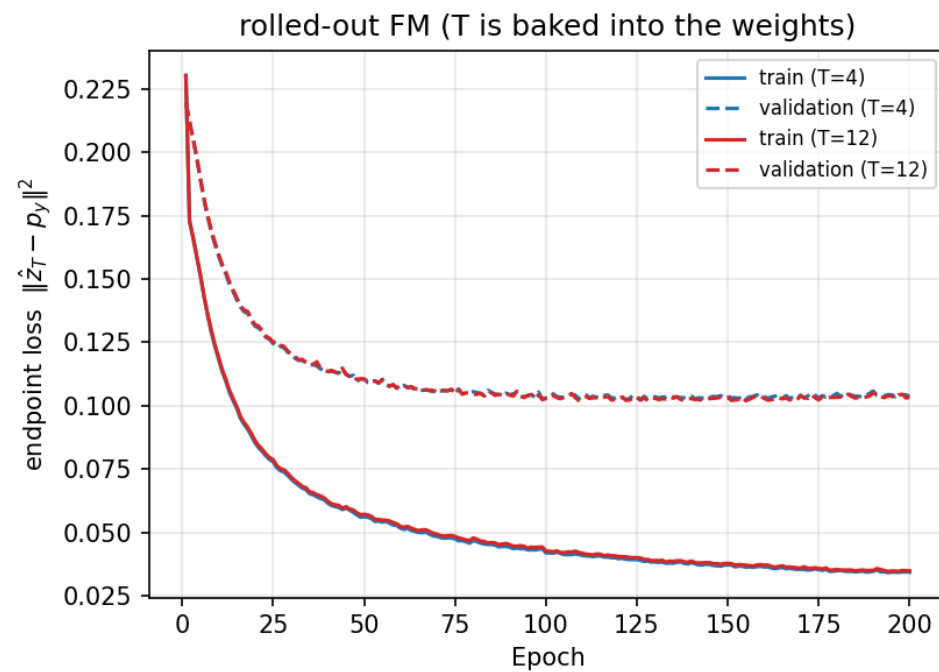
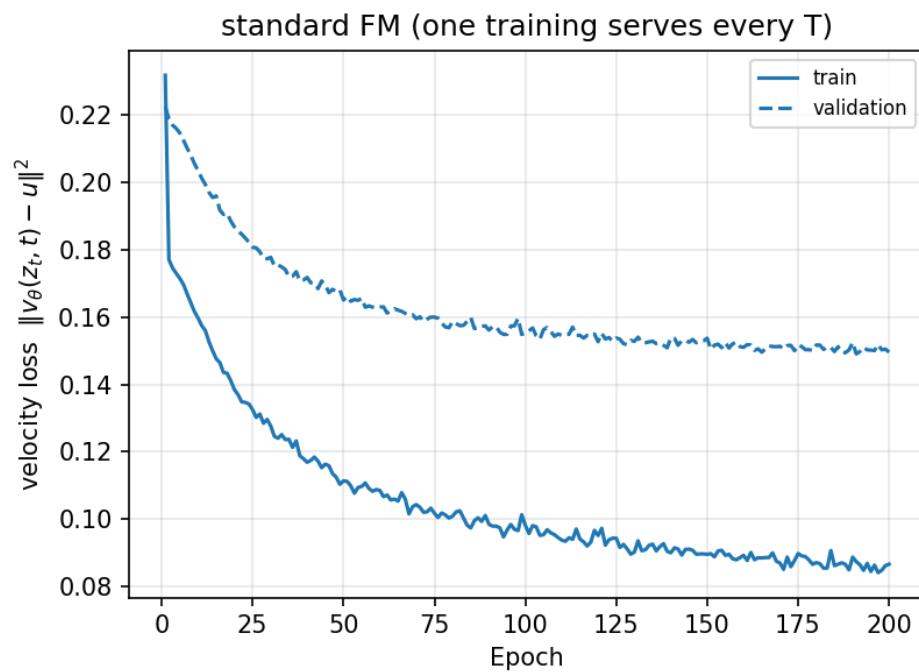
dtd / resnet18: flow-matching training (10-shot, seed 0)
note: the two panels measure different quantities and are not comparable to each other



Stage 2: flow-matching training curves (10-shot, seed 0)

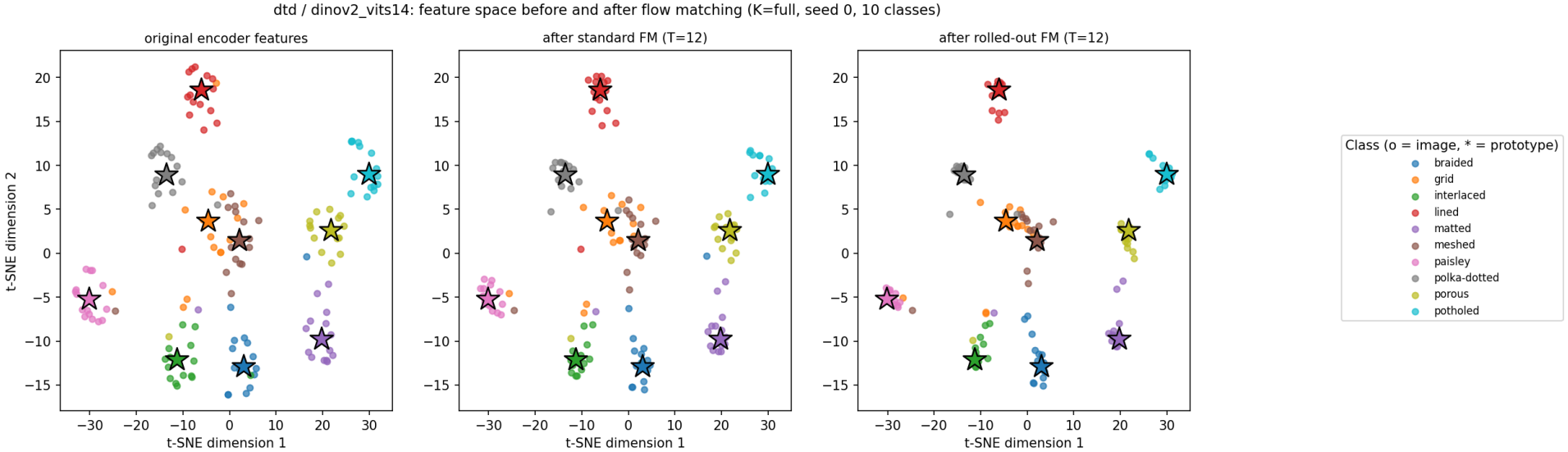
flowers102 / resnet18

flowers102 / resnet18: flow-matching training (10-shot, seed 0)
note: the two panels measure different quantities and are not comparable to each other



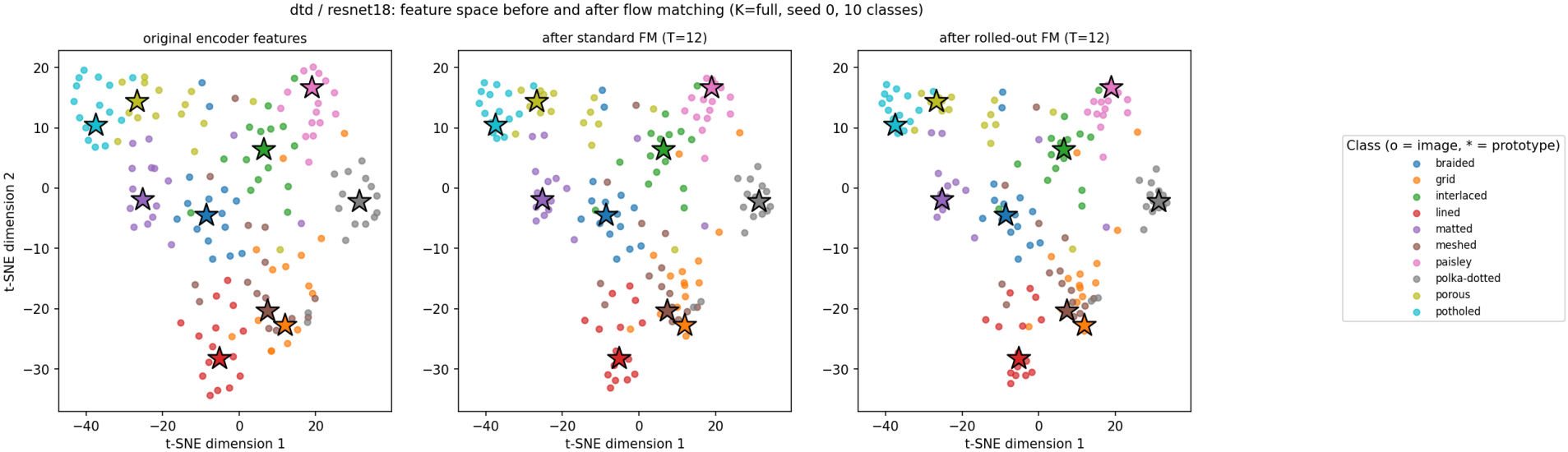
Stage 2: feature space before and after flow matching (t-SNE)

dtd / dinov2_vits14



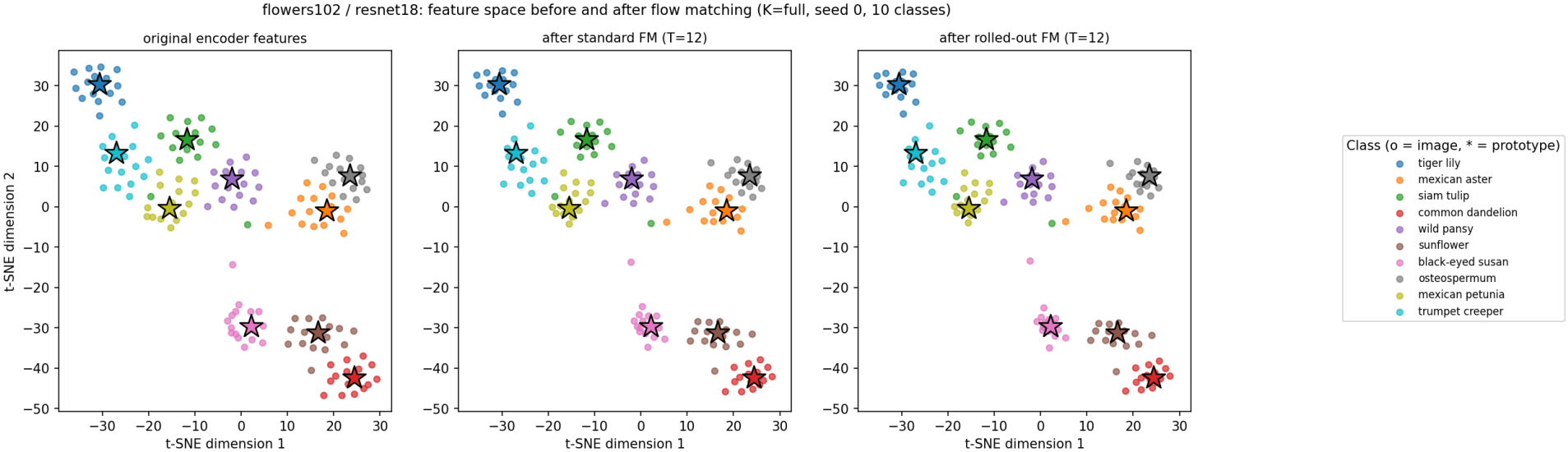
Stage 2: feature space before and after flow matching (t-SNE)

dtd / resnet18



Stage 2: feature space before and after flow matching (t-SNE)

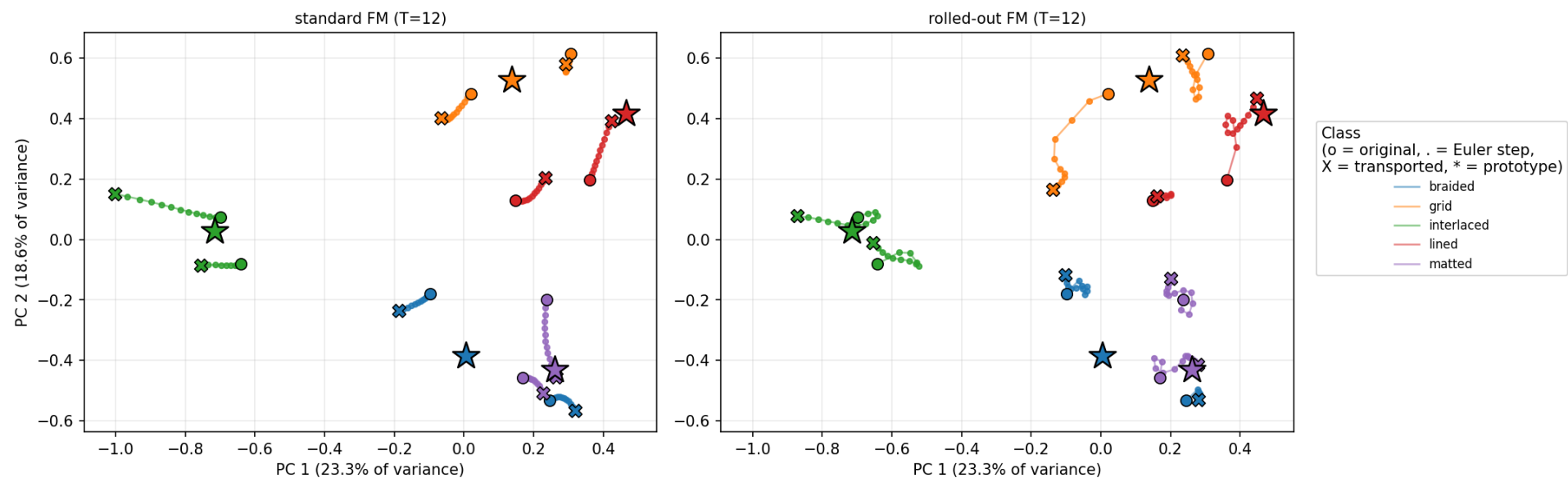
flowers102 / resnet18



Stage 2: flow trajectories (PCA)

dtd / dinov2_vits14

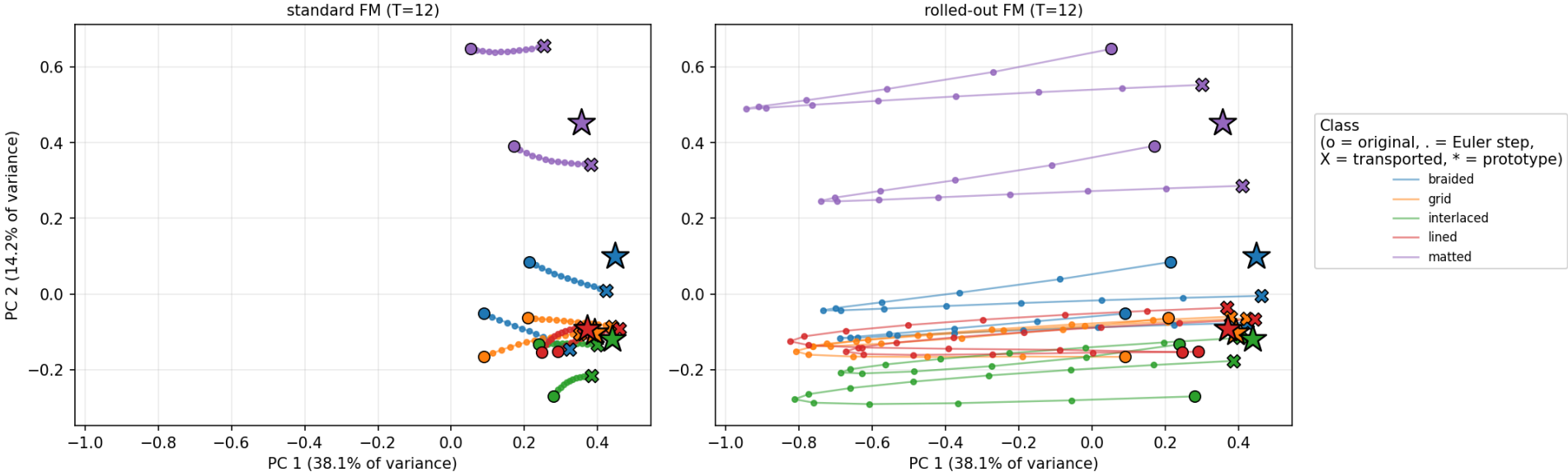
dtd / dinov2_vits14: Euler flow trajectories (K=full, seed 0, 10 test examples)



Stage 2: flow trajectories (PCA)

dtd / resnet18

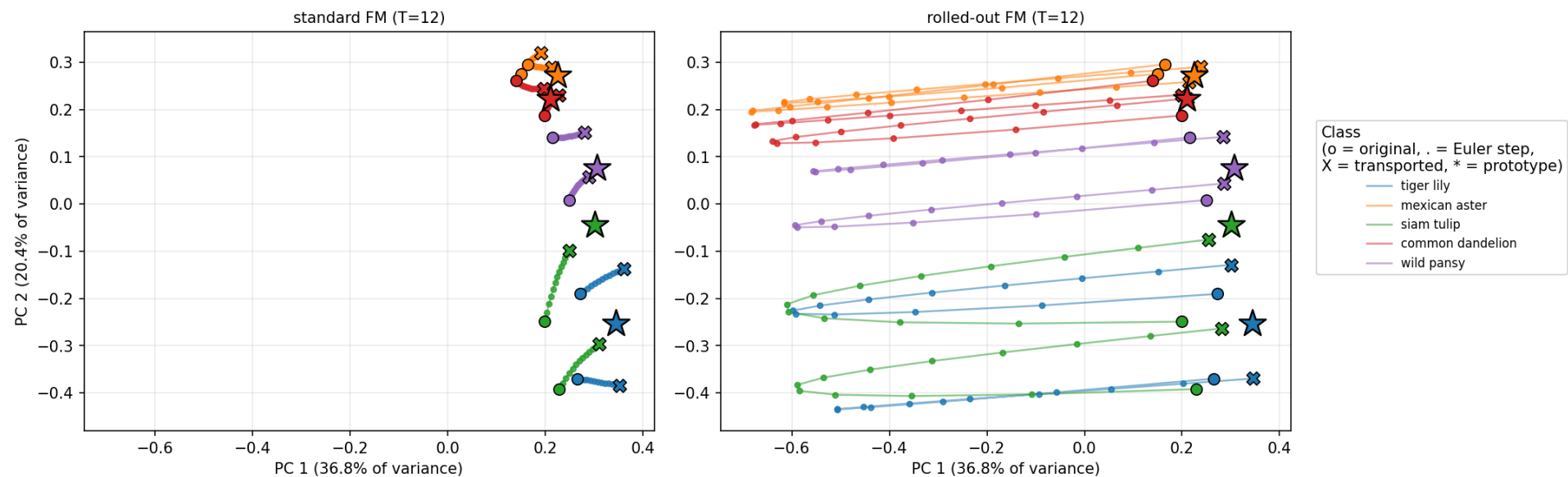
dtd / resnet18: Euler flow trajectories (K=full, seed 0, 10 test examples)



Stage 2: flow trajectories (PCA)

flowers102 / resnet18

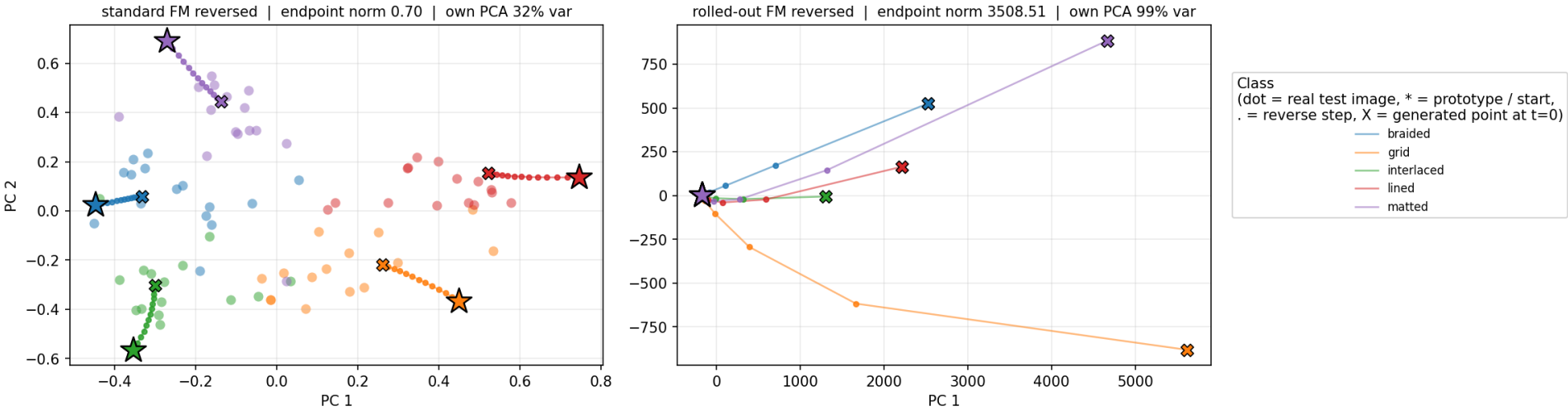
flowers102 / resnet18: Euler flow trajectories (K=full, seed 0, 10 test examples)



Stage 2: reverse flow from the class prototypes (PCA)

dtd / dinov2_vits14

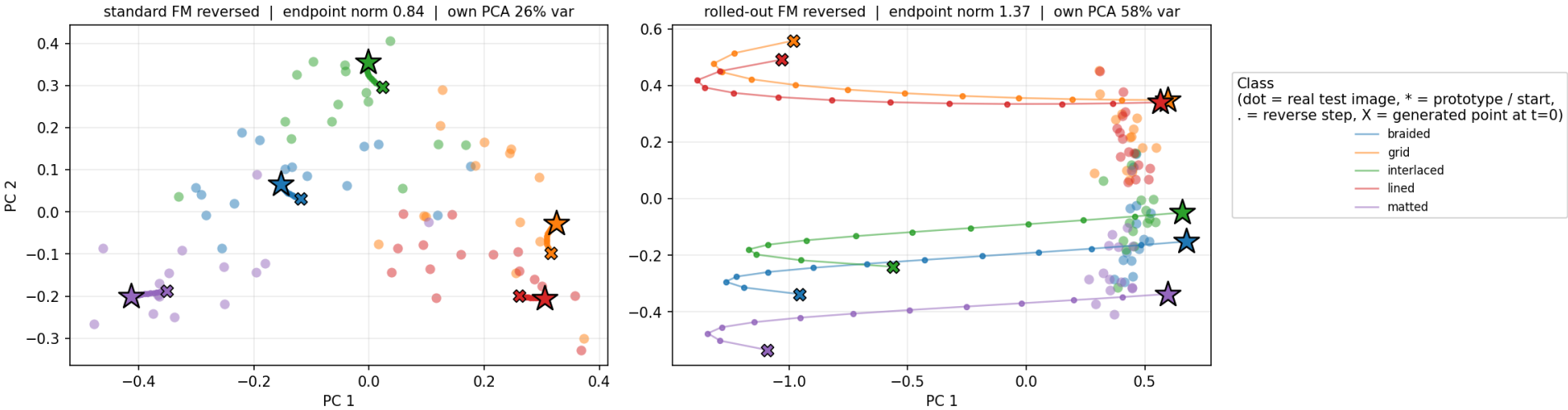
dtd / dinov2_vits14: reverse flow from the class prototypes (K=full, seed 0, T=12)
coloured dots are real test features of the same classes - does the generated point land among them?
each panel has its own PCA basis and scale - not comparable by eye; see endpoint norms



Stage 2: reverse flow from the class prototypes (PCA)

dtd / resnet18

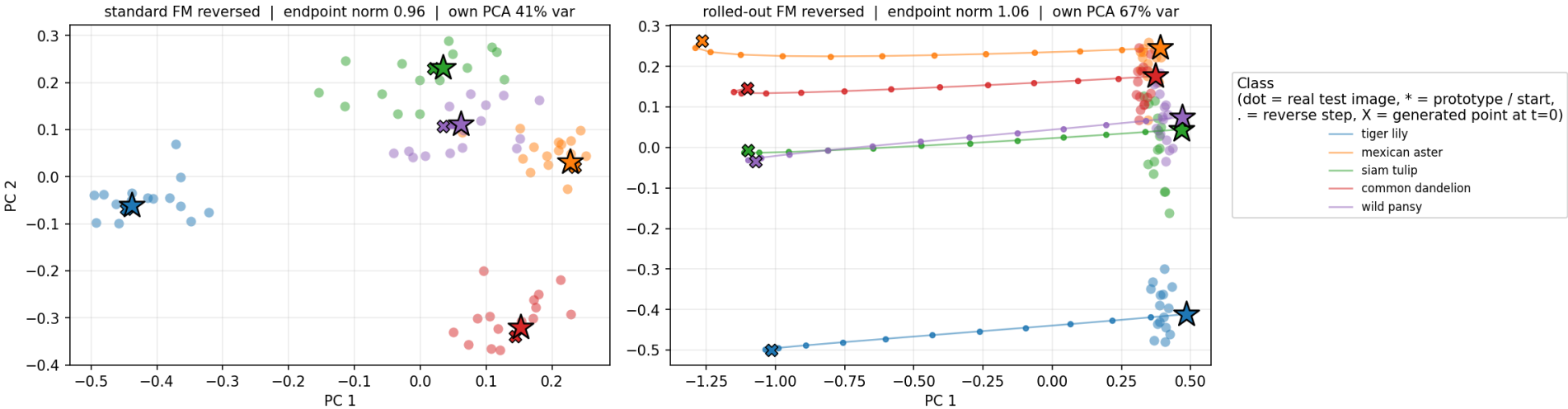
dtd / resnet18: reverse flow from the class prototypes (K=full, seed 0, T=12)
coloured dots are real test features of the same classes - does the generated point land among them?
each panel has its own PCA basis and scale - not comparable by eye; see endpoint norms



Stage 2: reverse flow from the class prototypes (PCA)

flowers102 / resnet18

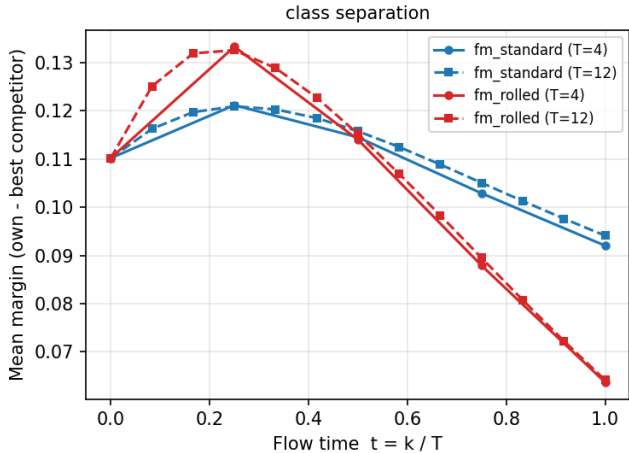
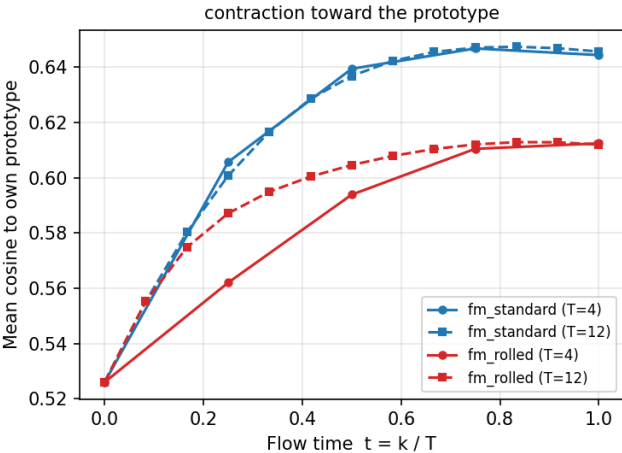
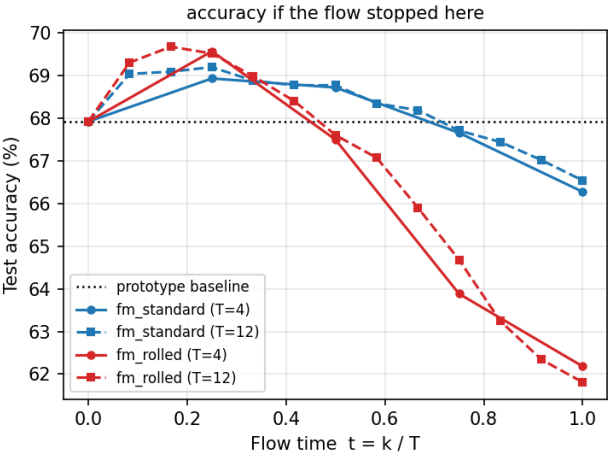
flowers102 / resnet18: reverse flow from the class prototypes (K=full, seed 0, T=12)
coloured dots are real test features of the same classes - does the generated point land among them?
each panel has its own PCA basis and scale - not comparable by eye; see endpoint norms



Stage 2: metrics along the flow

dtd / dinov2_vits14 (K=10)

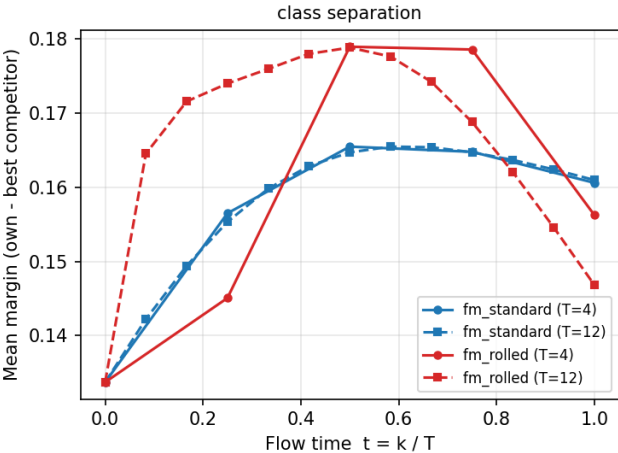
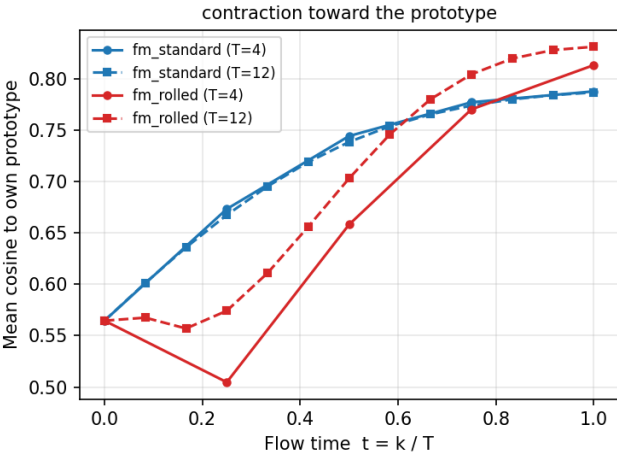
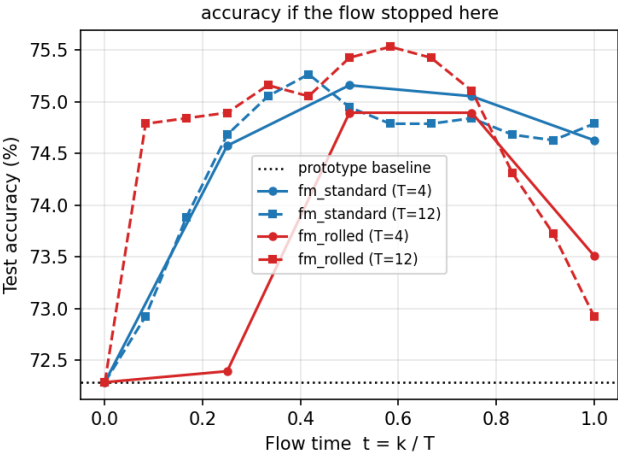
dtd / dinov2_vits14: metrics along the flow (K=10)



Stage 2: metrics along the flow

dtd / dinov2_vits14 (K=full)

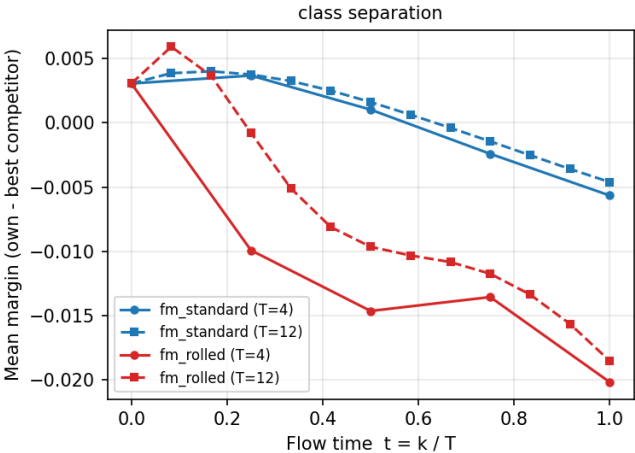
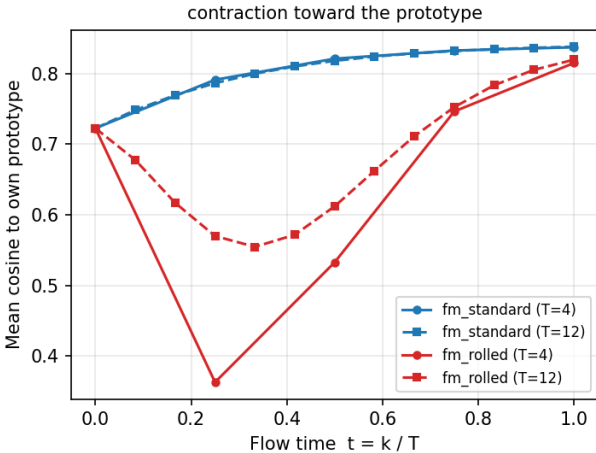
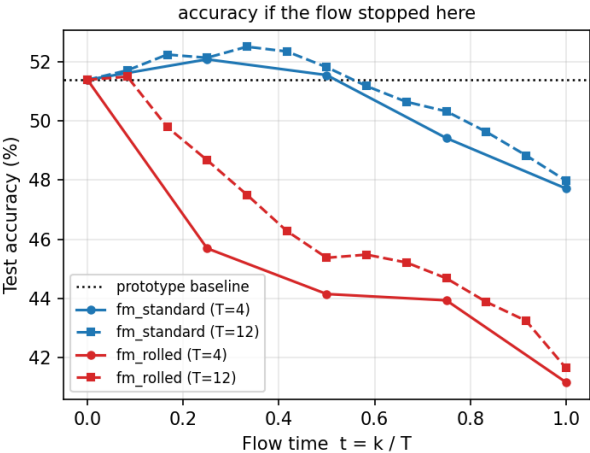
dtd / dinov2_vits14: metrics along the flow (K=full)



Stage 2: metrics along the flow

dtd / resnet18 (K=10)

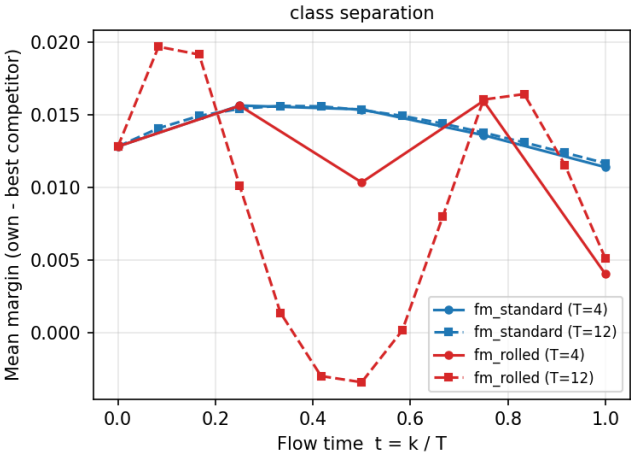
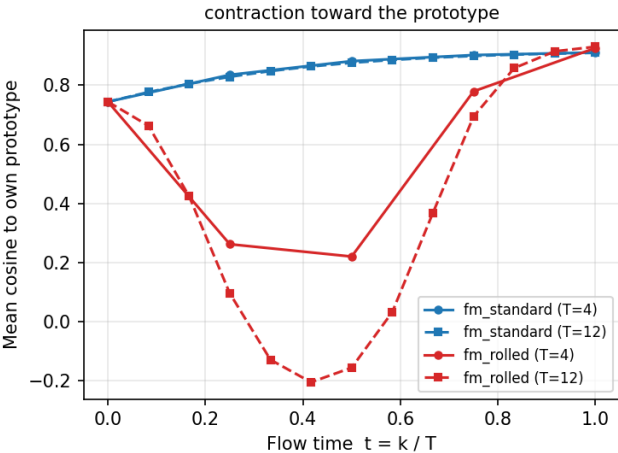
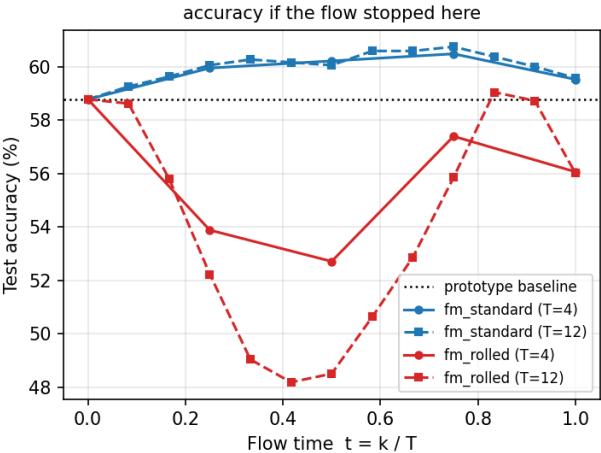
dtd / resnet18: metrics along the flow (K=10)



Stage 2: metrics along the flow

dtd / resnet18 (K=full)

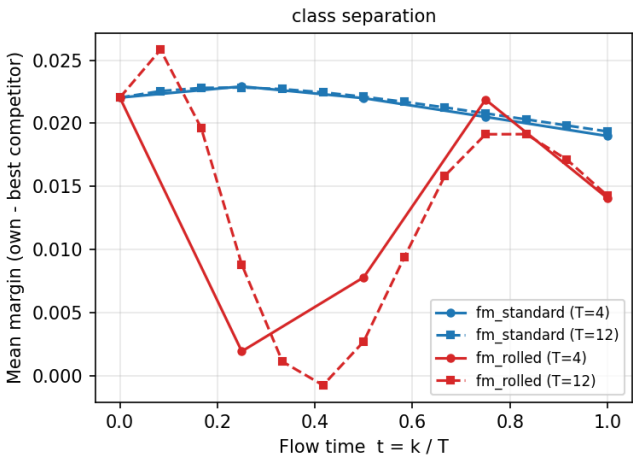
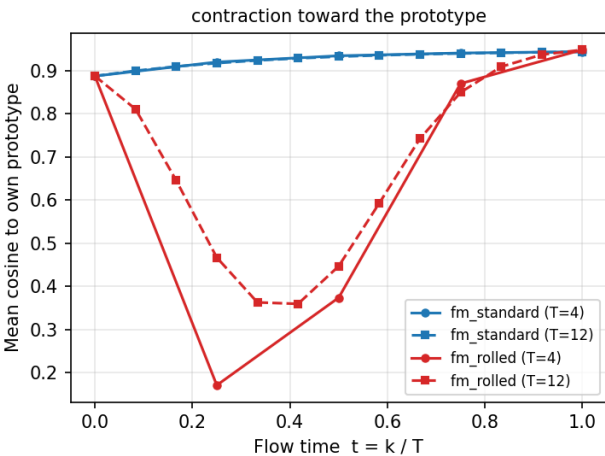
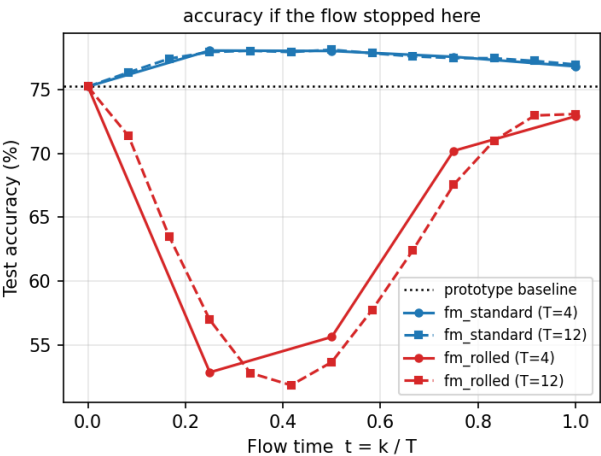
dtd / resnet18: metrics along the flow (K=full)



Stage 2: metrics along the flow

flowers102 / resnet18 (K=10)

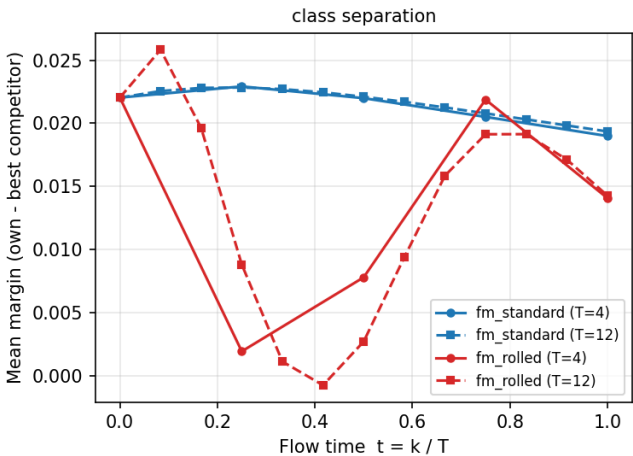
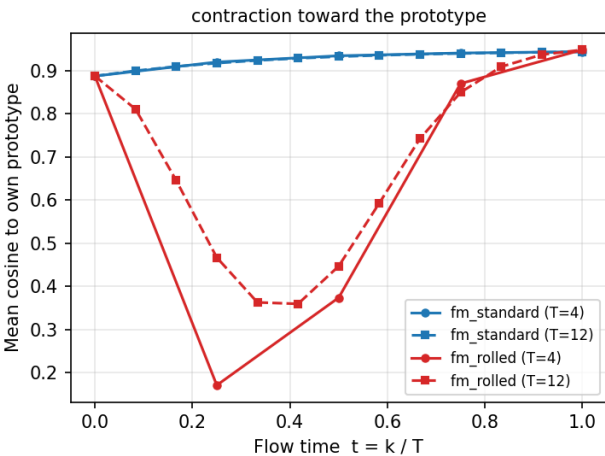
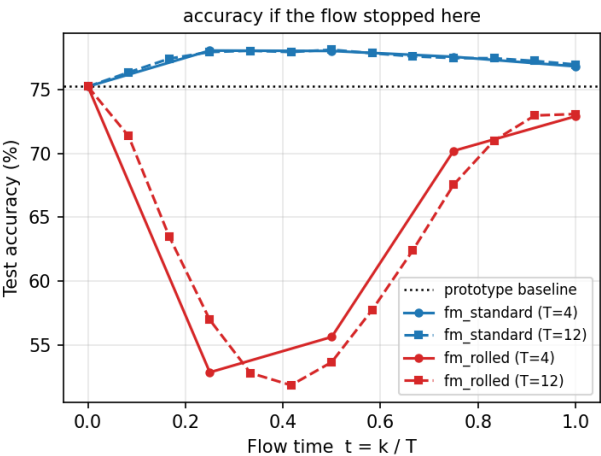
flowers102 / resnet18: metrics along the flow (K=10)



Stage 2: metrics along the flow

flowers102 / resnet18 (K=full)

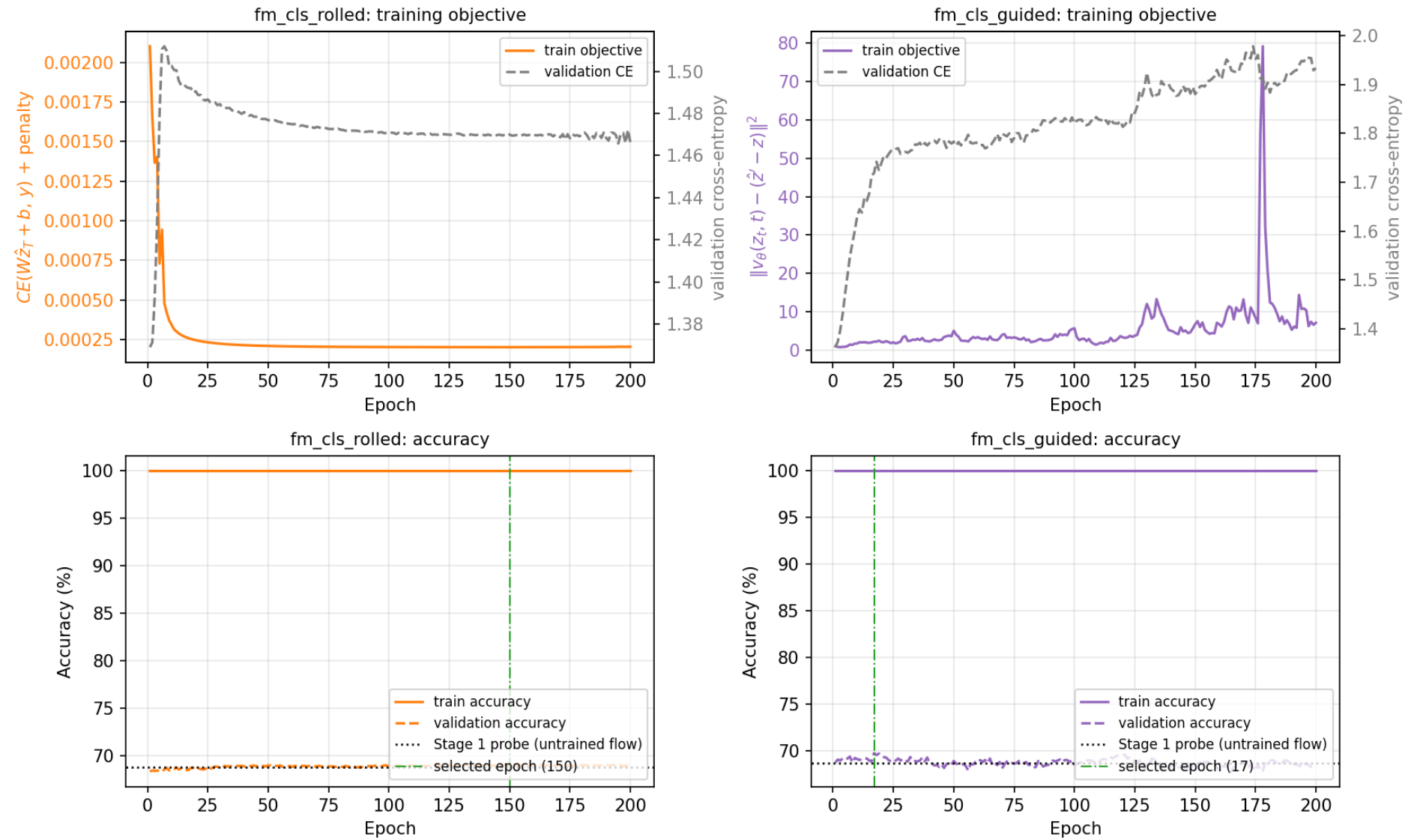
flowers102 / resnet18: metrics along the flow (K=full)



Stage 3: training curves (10-shot, seed 0)

dtd / dinov2_vits14

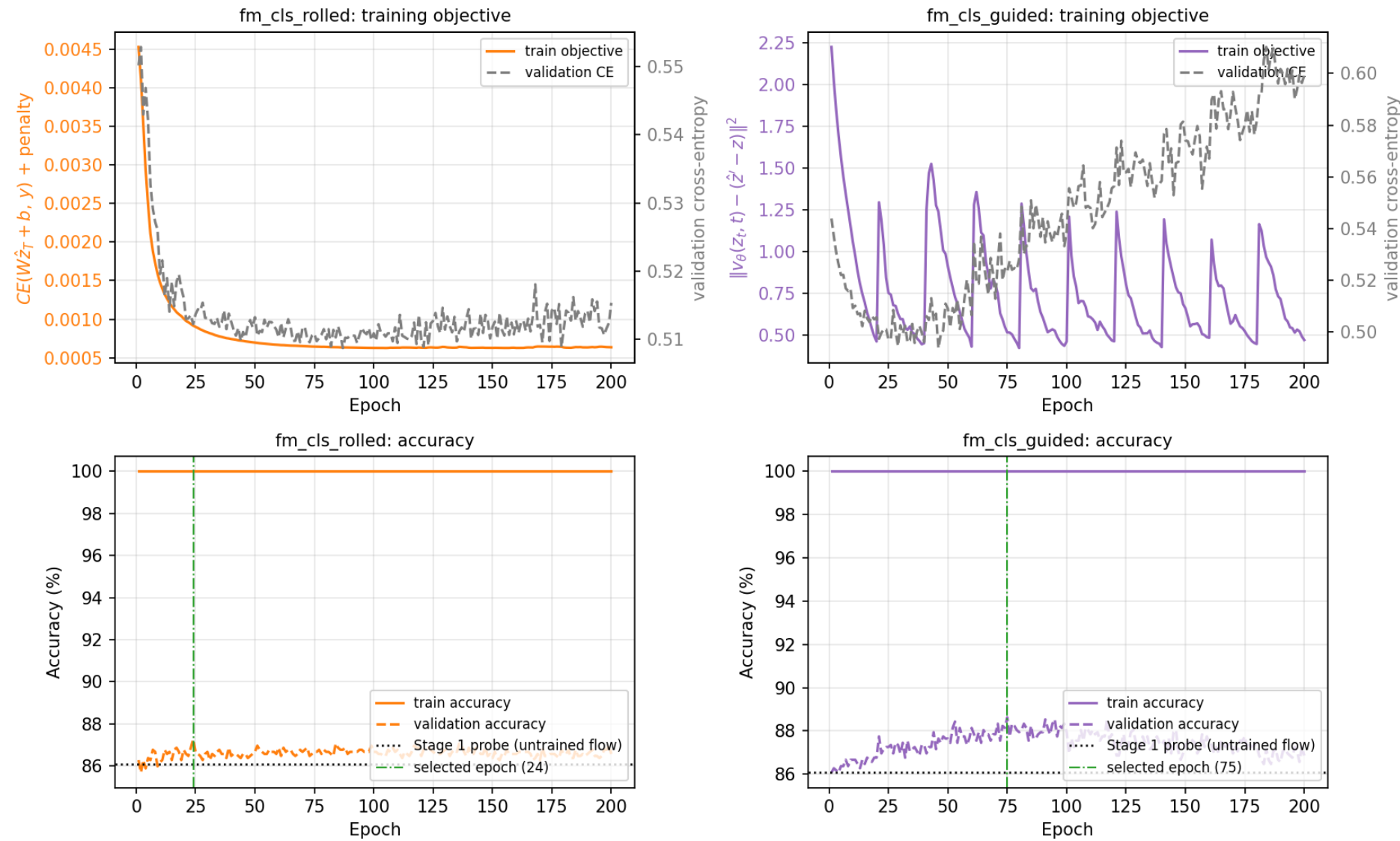
dtd / dinov2_vits14: Stage 3 training (10-shot, seed 0)
the two training objectives are different quantities and are not comparable to each other; the accuracy panels are



Stage 3: training curves (10-shot, seed 0)

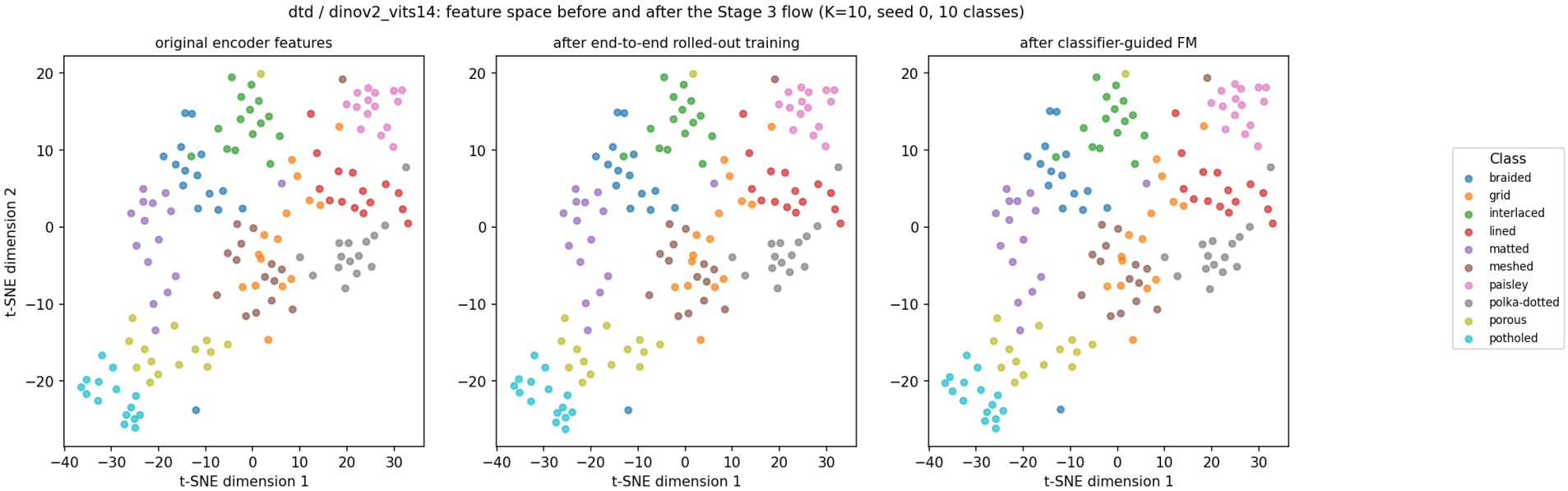
flowers102 / resnet18

flowers102 / resnet18: Stage 3 training (10-shot, seed 0)
the two training objectives are different quantities and are not comparable to each other; the accuracy panels are



Stage 3: feature space before and after the flow (t-SNE)

dtd / dinov2_vits14



Stage 3: feature space before and after the flow (t-SNE)

flowers102 / resnet18

